

SACS



Tow

Version 24.00

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TABLE OF CONTENTS

1	INTRODUCTION	4
1.1	OVERVIEW.....	4
1.2	PROGRAM FEATURES.....	4
2	TOW INPUT DATA.....	5
2.1	GENERAL OPTIONS.....	5
2.1.1	Excluding the Weight of the Model	5
2.2	STRUCTURE ORIENTATION.....	5
2.2.1	Repositioning the Structure	6
2.2.2	Center of Motion.....	6
2.2.3	Tow Local Coordinate System	6
2.3	CONVERTING MODEL LOADING TO WEIGHT	8
2.3.1	Designating Load Cases to be Converted	8
2.3.2	Converting the First Load Case	8
2.4	SPECIFYING USER DEFINED WEIGHTS	9
2.4.1	User Defined Weights and Seastate Data	9
2.5	GENERATING INERTIA LOAD CASES	9
2.5.1	Specifying Motion Data.....	10
2.5.2	Specifying Acceleration Data	10
2.5.3	Specifying Allowable Stress Modifier Inertia Load Cases	11
2.5.4	Alternative Motion Data	11
2.5.5	Alternative Acceleration Data	12
2.6	REPORTS.....	12
2.6.1	Joint Special Reports	12
2.7	GENERATING RIGID BODY MASS MATRIX.....	12
2.8	CALCULATING MOTION RESPONSE DUE TO RAO INPUT	12
2.8.1	RAO Input Data	13
2.8.2	Wave Definition Data	13
2.8.3	Automatic Load Case Generation from RAO Input Data	14
2.8.4	Alternative RAO Input	14
3	COMMENTARY	16
3.1	DETERMINING INERTIA FORCES.....	16
3.2	CREATING INERTIA LOAD	17
3.2.1	Modeled Elements	17
3.2.2	User Defined Weights and Loads	17
4	SAMPLE PROBLEMS.....	18
4.1	SAMPLE PROBLEM 1	20
4.2	SAMPLE PROBLEM 2	29
4.3	SAMPLE PROBLEM 3	36
5	INPUT LINES.....	42

1 INTRODUCTION

1.1 OVERVIEW

The Tow program module is used to generate forces typically encountered during barge transportation of structures. The program generates inertia forces due to acceleration of the structure, in the form of joint concentrated and member distributed loads.

1.2 PROGRAM FEATURES

Tow requires a SACS input model file and a Tow input file for execution. The program creates an output structural data file containing the model and load cases consisting of inertia loads generated by the program.

Some of the main features and capabilities of Tow program module are:

1. Creates load case for X, Y and/or Z translational and/or angular accelerations.
2. Calculates the mass of modeled beam and plate elements for inertia load generation automatically.
3. Generates member distributed loads and joint concentrated loads.
4. Supports joint concentrated, member concentrated and member distributed user defined weights to account for non-modeled items.
5. Allows user defined loads to be specified on LOAD input lines in the Tow input file.
6. Option to consider loading specified in the model file when creating inertia load.
7. Ability to orient or reorient the structure within the Tow input file for any particular inertia load case to be created.
8. Ability to convert any load case in the SACS input file to weights.
9. Allows factoring or any SACS load case when converting to weights.
10. Optionally considers the weight of grout.
11. Ability to calculate acceleration data based on user defined pitch, roll and/or yaw angle and period.
12. Generates rigid body mass matrix including the mass of members, plates and any added weights.
13. Ability to determine barge responses based on wave characteristics and RAOs.
14. Allows specification of allowable stress modifiers in the Tow input file.

Note: The sum of the moments in the listing file is about the C.G. of the structure.

2 TOW INPUT DATA

The Tow program requires a SACS model file and a Tow input file. The model file requires no special modeling consideration for the purpose of inertia load case generation.

2.1 GENERAL OPTIONS

Tow analysis options are specified on the TOWOPT line. Enter the input units in columns 9-10. Enter 'EC' in columns 11-12 for the input data to be echoed. If concentric tubular sections are to be considered grouted, enter 'GR' in columns 25-26 to include the weight of grout.

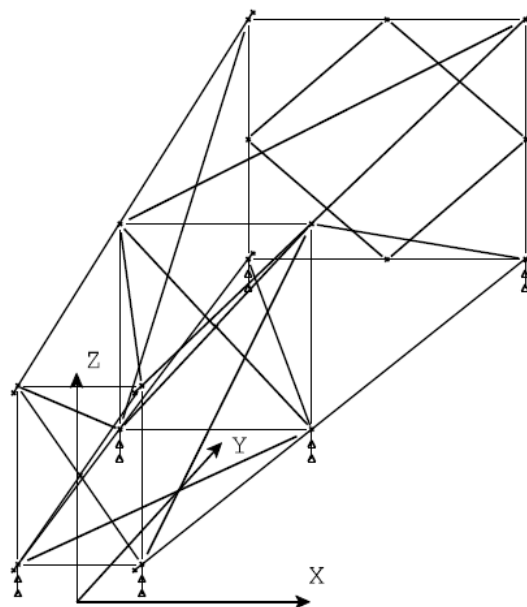
2.1.1 Excluding the Weight of the Model

By default, the weight of the modeled elements is included in the inertia loading generated. The weight of the model can be ignored, however, by inputting 'LO' in columns 27-28 on the TOWOPT line. Load cases generated would only contain inertia loading due to any load cases or user defined weights converted to inertia load.

2.2 STRUCTURE ORIENTATION

Normally the structure would be rotated to the proper orientation and position on the barge in the model file such that the global structural axes are aligned with the barge axes.

This allows the user to generate inertia loads due to surge, sway and heave linear accelerations and roll, pitch and yaw angular acceleration along and/or about the structural global axes. See the figure shown below.



Note: Inertia loads may be generated in any direction and are not required to be generated along the structural global axes. However, orienting the structure in the model so the global axes are aligned with the barge axes allows the user to verify and visualize transportation results quite easily.

2.2.1 Repositioning the Structure

By default, the position of the structure as modeled is used for the Tow analysis. The structure may be repositioned for any inertia load case created using the POSITION line. When orientation angles are specified, the structure is rotated about the global X axis first, then the global Y axis, followed by rotation about the global Z axis.

Note: When the structure is repositioned using the POSITION line, the new position is retained for all subsequent inertia load cases created until a new position is defined.

2.2.2 Center of Motion

The center of motion is the point about which the barge/structure system rotates when subject to transportation forces that induce pitch, roll and/or yaw. The X, Y and Z coordinates of this point are specified in the Tow input file on the TOWOPT input line and are defined with respect to the model global coordinate system. Enter the global X, Y and Z coordinates of the center of motion in columns 31-37, 38-44 and 45-51, respectively. The following designates that the center of motion is defined by the global X, Y and Z coordinates 8.0, 23.5, 13.67.

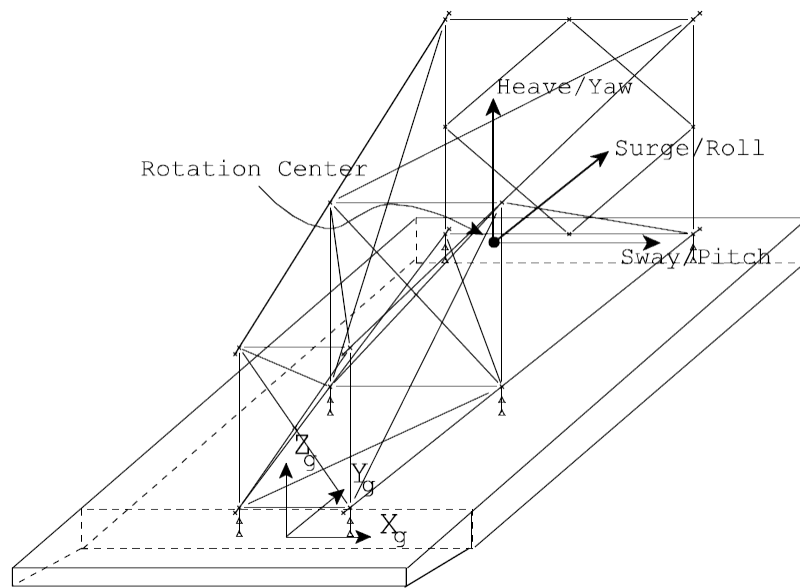
	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	TOW SAMPLE PROBLEM							
2	TOWOPT	EN		8.0	23.5	13.67		

2.2.3 Tow Local Coordinate System

The origin of the tow or transportation local coordinate system coincides with the center of motion specified on the TOWOPT input line. The orientation of the Tow coordinate system is defined by the user by specifying the model global axes that correspond to the surge/roll, sway/pitch and heave/yaw axes. Enter the structural global axis corresponding to the surge axis in column 52, the axis corresponding to the sway axis in column 53 and the axis corresponding to the heave axis in column 54.

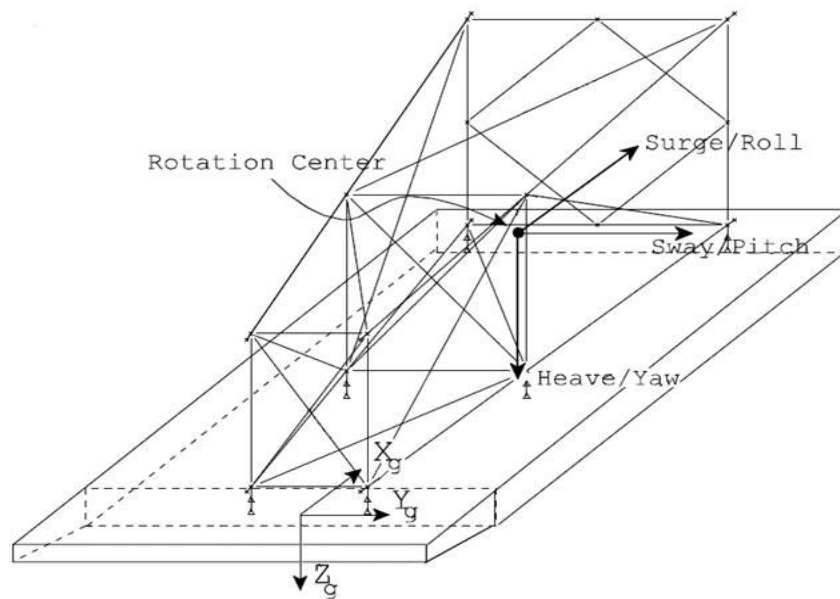
In the following figure, the global Y axis is the surge axis, the global X axis is the sway axis and the global Z is the heave axis as designated in columns 52-54.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	TOW SAMPLE PROBLEM							
2	TOWOPT	EN	OR	8.0	23.5	13.67	YXZ	



This figure is valid for YXZ (default), XZY, and ZYX tow coordinate systems.

For XYZ, YZX, and ZXY coordinate systems, the orientation of the coordinate system is switched to maintain a right-handed coordinate system. The following figure depicts the orientation of the XYZ tow coordinate system.



Note: Modification of the tow coordinate system is intended to accommodate motion data from external sources and is not recommended for reorienting the structure with respect to the tow coordinate system. Refer to the POSITION input line for repositioning of the structure.

2.3 CONVERTING MODEL LOADING TO WEIGHT

Inertia loads for non-modeled items defined by member concentrated, member distributed and joint concentrated loads specified in the model file can be converted to weights automatically by the program.

By default, when member distributed loads are converted to weight, they are treated as member distributed weights. Member distributed loads may be converted to joint weights by specifying 'ND' in columns 15-16 on the TOWOPT input line.

2.3.1 Designating Load Cases to be Converted

Load cases in the model file that are to be converted to weight are designated using the LCFAC line.

The load cases may be factored when converting to weight by specifying the appropriate factor on the LCFAC line. Enter the load case factor in columns 11-16 and any load cases to which this factor is to be applied in columns 17-75.

Note: The LCFAC line should precede any ACCL/ACCEL or MOTION/MOTN lines defining inertia load cases that are to include these weights.

In the following example, 100% of loads cases 'EQP1' and 'EQP2' are to be converted to weight when generating the inertia load cases 'R+H1' and 'R-H1' defined by the first two MOTION lines, while 50% of load cases 'EQP3' and 'EQP4' are to be used when generating load case 'R+H2' defined by the third MOTION line.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	LCFAC	1.0EQP1	EQP2					
2	MOTIONR+H1	12.0	10.0		0.20			G
3	MOTIONR-H1	12.0	10.0		-0.20			G
4	LCFAC	0.5EQP3	EQP4					
5	MOTIONR+H2	12.0	12.0		0.20			G

Note: Load cases selected to convert to weight replace any previous load cases selected and are used for all subsequent inertia load cases created by either MOTION or ACCL lines until other load cases are selected. In the above sample, only load cases 'EQP3' and 'EQP4' were used for load case 'R+H2'.

2.3.2 Converting the First Load Case

If the model file contains only one load case, it may be converted to weight by specifying 'LD' in columns 13-14 on the TOWOPT line.

Note: The 'LD' option is provided for backward compatibility purposes only and should not be used in conjunction with the LCFAC line.

2.4 SPECIFYING USER DEFINED WEIGHTS

Additional weights may also be specified in the Tow input file using Joint Weight and Member Weight input lines. In order to include user defined weight, the WEIGHT input line(s) should be specified prior to the MOTION or ACCL input line used to define the inertia load cases to be created. The specified weights will be included for all subsequent load cases created unless overridden by new WEIGHT input lines.

Note: When specifying weight data, all weight data defined for previous inertia load cases is replaced. Therefore, all weight data may be removed for an inertia load case by specifying a 0.0 weight.

For example, the following input file has user defined joint and member weights specified. A load of 20.0 at joint 599 defined by the WEIGHT input line will be included in the first inertia load case along with a concentrated weight of 20.0 on member 506-606. For the second and all subsequent load cases, all weight data is removed.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WEIGHT	599		20.0				
2	WEIGHT	506	606	20.0			5.0	
3	ACCL		1.0					
4	WEIGHT	599		0.0				
5	ACCL			1.0				

Note: Member concentrated, member distributed and joint concentrated LOAD input lines can be entered directly into the TOW input file after the POSITION input line (or TOWOPT line if no POSITION input line is specified). No modifications to the LOAD input lines are required. The program will automatically create the corresponding weights.

2.4.1 User Defined Weights and Seastate Data

The Tow program can use weight groups defined in a Seastate input file to specify inertia loading. These weight groups are specified on the INCWGT line. For example, the following line indicates that weight groups 2, 3 and 5 will be used to create inertia load cases. These weight groups must be defined in the Seastate input file.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	INCWGT	2	3	5				

2.5 GENERATING INERTIA LOAD CASES

Load cases containing inertia loads due to translational linear acceleration and/or rotational angular acceleration about the global axes are created based on either motions or accelerations defined using either the MOTION or the ACCL input lines, respectively.

Inertia loading is generated automatically for any beam or plate element in the model based on its modeled weight. In addition, inertia load is developed for any loading converted to weight and user defined weights.

Inertia load is created in the opposite direction of the imposed acceleration (D'Alembert forces) and are calculated based on the structures current position (ie. the latest POSITION line encountered), the current selected load cases and factors (ie. the latest LCFAC data specified) and any LOAD and/or WEIGHT input lines defined prior to the MOTION or ACCL line defining the accelerations.

2.5.1 Specifying Motion Data

The Tow program can create inertia load cases based on accelerations defined by motion data input by the user on the MOTION line.

A unique load case is created for each MOTION line. Specify the load case name in columns 7-10.

Roll angle and period are defined in columns 11-15 and 16-20, pitch angle and period in columns 21-25 and 26-30 and yaw angle and period in columns 31-35 and 36-40.

When the barge is rotated by a roll or pitch angle q , the weight of the structure applied vertically down contains a component normal to the barge calculated as $\cos q$ and a component parallel to the barge surface calculated as $\sin q$. The Tow program can automatically include these effects of gravity on the translations inertial forces with the "Include Gravity Option" in column 77.

If heave is assumed normal to barge:

Enter 'G' in column 77 to include the gravity component for surge, sway, and heave.

Enter 'L' in column 77 to include gravity component for only surge and sway (lateral forces only).

If heave is assumed vertical:

Enter 'H' in column 77 to include the gravity component for surge, sway, and heave.

Enter 'V' in column 77 to include gravity component for only surge and sway (lateral forces only).

Note: Leave column 77 blank to exclude the effects of gravity or if the translational acceleration input already includes the appropriate gravity component.

The center of motion used for this inertia load case may be overridden in columns 56-76. The inertia due to structural weight may be ignored by specifying 'N' in column 78. This is especially useful if the MOTION load case will be combined with DEAD load cases. This way the structural weight will not be included twice (or more) in the load combination.

The following defines inertia load cases 'R+H1' and 'R-H1'. The roll angle is 15.0 degrees with a period of 10.0 seconds. The first load case has a heave acceleration of +0.2 while the second has a heave acceleration of -0.20. Both have the program include the effects of gravity automatically.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	MOTIONR+H1	15.0 10.0			0.20			G
2	MOTIONR-H1	15.0 10.0			-0.20			G

2.5.2 Specifying Acceleration Data

The ACCL line may be used to specify accelerations used to generate the inertia load case. A unique inertia load case is created for each ACCL line specified.

Enter the name of the load case in columns 7-10, unless the name of the load case is to be determined by the program based on the next available numeric load case name. The applicable X, Y and/or Z axis angular acceleration(s) are specified in deg/sec² units in columns 11-20, 21-30 and 31-40, respectively. The translational or linear X, Y and Z axis accelerations are expressed in G's in columns 41-50, 51-60 and 61-70, respectively. For example, the following creates a load case containing inertia loading due to a linear acceleration along the global X axis of 1G.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	ACCL				1.0			

Note: The effect of gravity is not automatically included when generating inertia loads using the ACCL line. Gravity load may be created by specifying 1.0 G acceleration in the +Z direction.

2.5.3 Specifying Allowable Stress Modifier Inertia Load Cases

Allowable stress modifier may be specified for any inertia load case created using the AMOD line. Specify the load case name and the appropriate factor. For example, the following creates inertia load cases R+H1 and R-H1, and allowable stress factor of 1.33 is applied to both load cases.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	TOWOPT	EN		8.0	23.5	13.67		
2	AMOD							
3	AMOD	R+H1	1.33	R-H1	1.33			
4	MOTIONR+H1	15.0	10.0		0.20			G
5	MOTIONR-H1	15.0	10.0		-0.20			G

Note: The AMOD line must contain a header line and must follow directly after the TOWOPT line.

2.5.4 Alternative Motion Data

The MOTN line provides an alternative to the MOTION line in creating inertia load cases. The MOTN line is based on the inertia capabilities of the Seastate program. As such, a Seastate input file is needed to make use of its full capabilities. In the MOTN line, the load case name is specified in columns 7-10. The roll, pitch and yaw angles and periods are specified in columns 11-52. The translational acceleration components are specified in columns 53-73. The effects of gravity are included automatically with 'G' in column 74; structural weight is excluded from the load case with 'N' in column 75. All these options are similar to the MOTION line. The MOTN line has the added feature of specifying a center of motion for rotational accelerations and velocities. This center identifier is specified in columns 77-80. This center must be defined in the Seastate input file.

The following defines inertia load cases cases 'R+H1' and 'R-H1'. The roll angle is 15.0 degrees with a period of 10.0 seconds. The first load case has a heave acceleration of +0.2 while the second has a heave acceleration of -0.20. Both have the program include the effects of gravity automatically and neither excludes structural weight. The center ID for these load cases, which is specified in the Seastate input file, is 'CEN2'.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	MOTN	R+H1	15.0	10.0			0.2G	CEN2
2	MOTN	R-H1	15.0	10.0			-0.20G	CEN2

2.5.5 Alternative Acceleration Data

The ACCEL line provides an alternative to the ACCL line in specifying accelerations to create inertia load cases. The ACCEL line is based on the inertia capabilities of the Seastate program. As such, a Seastate input file is needed to make use of its full capabilities. In the ACCEL line, the load case name is specified in columns 6-9. The translational acceleration components are specified in columns 10-30. The rotational accelerations are specified in columns 31-51 with the rotational velocities specified in columns 52-72. The effects of gravity are included automatically with 'G' in column 75. The ACCEL line offers two added features when compared with the ACCL line: 1) the specification of angular velocities for the computation of centripetal accelerations, and 2) the specification of a center of motion for rotational accelerations and velocities. This center identifier is specified in columns 77-80. This center must be defined in the Seastate input file.

2.6 REPORTS

Output reports including member data, plate data, added weight and load summation are designated on the TOWOPT line.

Enter 'MP' in columns 17-18 to print a member data report and/or 'PP' in columns 19-20 to print a plate data report containing the weight of each member or plate. Enter 'WP' in columns 21-22 for a detailed report of user defined added weights. Inertia load summation reports may be generated about the model origin, center of motion or center of gravity by specifying either 'OR', 'RC' or 'GR', in columns 23-24.

2.6.1 Joint Special Reports

The location and acceleration for up to sixteen joints can be reported for each inertia load case created. The location and acceleration are reported in both the model global and Tow local coordinate systems. The joint names are specified in the Tow input file on the JTNUM input line.

2.7 GENERATING RIGID BODY MASS MATRIX

The program can generate a rigid body mass matrix including the mass of all members and plates included in the model. Any load cases converted to weights in addition to any weights or loads defined by the user in the input file are also included.

The following designates that the rigid body mass matrix be generated for the model including the mass due to load cases 'EQP1' and 'EQP2' specified on the LCFAC line.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	TOWOPT	EN		8.0	23.5	13.67		
3	LCFAC		EQP1 EQP2					
3	RBMASS							

2.8 CALCULATING MOTION RESPONSE DUE TO RAO INPUT

The Tow program can calculate the inertia response loading for various points along a wave cycle based on Response Amplitude Operator data input by the user.

Typically, for a particular wave direction, acceleration, velocity or displacement response operators are designated for various wave periods using RAO lines. From this data, inertia loading can be generated for various phase angles of the desired waves designated on the WAVDEF lines.

Note: A set of RAO and WAVDEF lines is designated for each wave direction when considering multiple wave directions using different response operators.

The loading due to a designated wave is assumed to be from a single amplitude response, where the wave height is assumed to be double amplitude. For example, a unit displacement RAO would yield a displacement equal to one half of the wave height.

2.8.1 RAO Input Data

Response Amplitude Operator data is input using a series of RAO lines for the particular wave direction in question.

Each RAO line is used to define the response for a particular frequency or period. Enter the amplitude type in column 5 as 'A' for acceleration, 'V' for velocity, 'D' for displacement or 'G' for acceleration in G's. Enter 'P' and the period or 'F' and the frequency in column 6 and columns 7-12, respectively. Designate the surge, sway, heave, roll, pitch and yaw amplitude and the phase angle corresponding to this frequency or period.

Note: One RAO line is required for each frequency or period used to define the response.

For example, the following input shows RAO data for wave periods of 25 and 20 seconds.

		1	2	3	4	5	6	7	8		
		12345678901234567890123456789012345678901234567890123456789012345678901234567890									
1	RAO GP	25.0	.938	137.	.001-43.0	.980	62.0	.004133.0	.430-47.0	.001	133.
2	RAO GP	20.0	.680	156.	.002-29.0	.954	87.0	.010149.0	.574-30.0	.001	150.

2.8.2 Wave Definition Data

WAVDEF lines are used to designate the waves for which responses are to be determined.

For each wave, the wave height and period must be specified in columns 15-20 and 21-26, respectively. The number of points per wave cycle to create loading is specified in columns 12-14 while the load case name of the first load case to be created is input in columns 7-10. Enter the wave direction (used for labeling only) in columns 27-32. If structural weight is to be excluded from the load case generated, specify 'N' in column 75. This is useful if the load case generated is combined with DEAD load cases. This way the structural weight will not be included twice (or more) in the load case.

The RAO data is interpolated for periods not defined by an RAO line. However, RAO data should not be extrapolated for waves with periods higher than the highest RAO period or lower than the lowest RAO period.

Note: A RAO line must be added with a period higher than the wave period if the wave period is greater than the highest RAO period. Likewise, a RAO line must be added with a period lower than the wave period if the wave period is less than the lowest RAO period.

The following specifies that response load cases for 18 points of a wave with height of 30 and period of 22 are to be generated. The wave direction is 45 degrees and the first load case is to be named 'A01'. A second 45 degree wave with a period of 20.0 is also specified. The first load is to be named 'B01'.

Load cases due to response from a set of 90 degree waves are also indicated by the second set of RAO and WAVDEF lines. None of the load cases defined will exclude structural weight.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	RAO GP	25.0	.938	137.	.001-43.0	.980	62.0	.004133.0
2	RAO GP	20.0	.680	156.	.002-29.0	.954	87.0	.010149.0
3	WAVDEF A01	18	30.0	22.0	45.			
4	WAVDEF B01	18	30.0	20.0	45.			
5	RAO GP	25.0	.950	180.	.005-83.0	.879	12.0	.006
6	RAO GP	20.0	.690	46.	.004-99.0	.956	67.0	.111104.0
7	WAVDEF C01	18	30.0	22.0	90.			
8	WAVDEF D01	18	30.0	20.0	90.			

Note: Interpolation will be used for the first wave in each direction since no RAO period corresponds to 22.0 seconds.

2.8.3 Automatic Load Case Generation from RAO Input Data

Load case data is automatically generated based on RAO input data by using the LCRAO line. The LCRAO line is used to define RAO characteristics to be used with the Response Amplitude Operators to calculate responses. It is normally used in tow fatigue analysis to generate the transfer function of "hot spot" stresses. For each period/frequency in all RAO lines, the LCRAO line will generate load cases starting with the load case specified in columns 7-10 of the line of the LCRAO line. If columns 7-10 are left blank, load cases are generated automatically and numbered consecutively in order of creation starting from 1. The wave direction in degrees is specified in columns 11-16 of the line.

The LCRAO line gives two options for load generation, 'R+I' and 'NPT', as specified in columns 17-19. The 'R+I' option will generate two load cases, real and imaginary, for each period/frequency in the RAO's. The two load cases are to be used in fatigue analysis with the 'R+I' option specified in the FTLOAD data records. The 'NPT' option will generate the number of load cases specified in columns 20-22 to complete each cycle of each period/frequency in the RAO's.

The following input shows RAO data for wave periods of 25 and 20 seconds. Two load cases, real and imaginary, will be generated per RAO with a wave direction of 45 degrees. These load cases are to be used in fatigue analysis with 'R+I' specified in the FTLOAD data records.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	RAO GP	25.0	.938	137.	.001-43.0	.980	62.0	.004133.0
2	RAO GP	20.0	.680	156.	.002-29.0	.954	87.0	.010149.0
3	LCRAO		45.0	R+I				

2.8.4 Alternative RAO Input

Another method of entering RAO data is supplied with the INCRAO line. This line allows the user to input RAO data from the Seastate input file into the program. Unlike other input methods for RAO's, the INCRAO offers the capability of specifying velocity-dependent RAO's. The velocity for the RAO is specified in columns 8-14. If English units are used for input, velocities may be specified in knots or feet

per second. If specified as feet per second, enter 'FPS' in columns 15-17. The following example will generate inertia loading for a wave speed of 2.5 feet per second.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	INCRAO	2.5FPS						

Note: This line is only used for RAO's defined in the Seastate input file. If velocity values specified in the INCRAO line do not correspond with velocity values in a Seastate RAO, interpolation or extrapolation will be used to obtain values.

3 COMMENTARY

3.1 DETERMINING INERTIA FORCES

The Tow program module generates reverse effect or D'Alembert inertia forces in the opposite direction of applied translational and rotational accelerations.

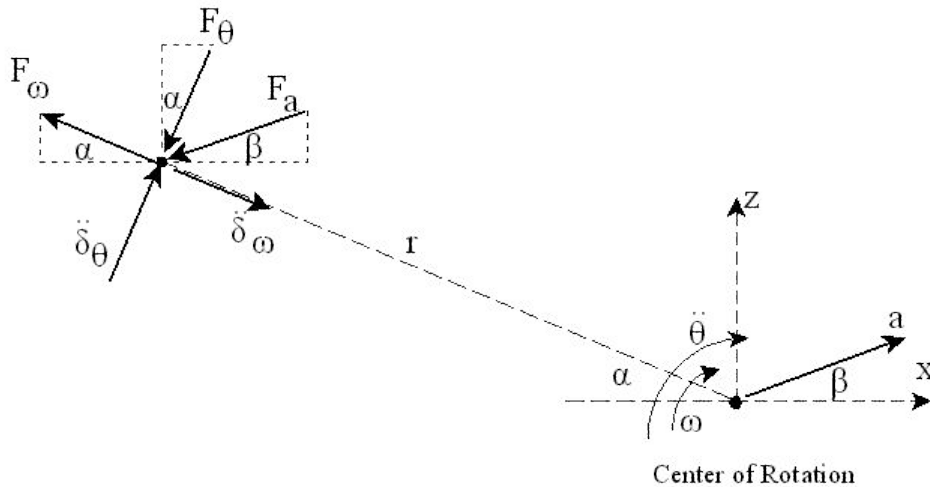
In the figure below, for any point on the structure with mass m , the inertia force, F_a , due to a translational/linear acceleration vector a , is determined by:

$$F_a = -ma$$

The force component acting along an axis x , F_{a_x} , is

$$F_{a_x} = -ma_x \quad (1)$$

where a_x is the component of the acceleration vector a along the axis ($a_x = a \cos \beta$).



The tangential acceleration δ_θ and the centrifugal acceleration δ_ω due to the rotation of the structure, may be taken as follows:

$$\delta_\theta = r\ddot{\theta} \quad \delta_\omega = r\omega^2$$

where ω and $\ddot{\theta}$ are the angular velocity and angular acceleration about the center of motion (center of rotation). The tangential and centrifugal forces are:

$$F_\theta = -m(r\ddot{\theta}) \quad F_\omega = -m(\omega^2 r)$$

The components along axis x of the rotational inertia forces F_θ and F_ω are:

$$F_{\theta_x} = -m(r\ddot{\theta} \sin \alpha) \quad (2)$$

$$F_{\omega_x} = -m(\omega^2 r \cos \alpha) \quad (3)$$

Combining equations 1, 2 and 3 yields the total inertia force along axis x, F_x , due to translational and rotational accelerations.

$$F_x = -m(a_x + r\ddot{\theta} \sin \alpha + \omega^2 r \cos \alpha)$$

If the angular velocity is negligible, $\omega=0$, the total inertia force along axis x is as follows:

$$F_x = -m(a_x + r\ddot{\theta} \sin \alpha) \quad (4)$$

Note: The TOW program assumes that the mass, m, is positive mass and equal for all directions.

3.2 CREATING INERTIA LOAD

3.2.1 Modeled Elements

For a beam element, mass m in equation 4 is the mass per unit length. For each segment of the element, a member uniform inertia load is created along the length of the segment based on the mass per unit length and the average acceleration of that segment.

Plate inertia load is calculated based on the plate total mass and the acceleration at the plate centroid. The total inertia load is applied at the plate corner joints as joint loads.

3.2.2 User Defined Weights and Loads

A uniform inertia member load is created for a user defined member distributed load or weight. The specified load or weight is converted into a positive distributed mass regardless of the load direction or the sign specified.

A inertia concentrated load is created for a user defined member concentrated load or weight. The specified load or weight is converted into a positive mass regardless of the direction of the load or the sign of the weight.

For a user defined joint weight and for user specified loads on a joint, an inertia joint load is created. Specified weight is converted into positive mass regardless of sign. For joint specified loads, the mass is determined from the vector sum of the x, y and z direction loads on the joint, P_x , P_y and P_z respectively, such that:

$$m = \frac{\sqrt{P_x^2 + P_y^2 + P_z^2}}{g}$$

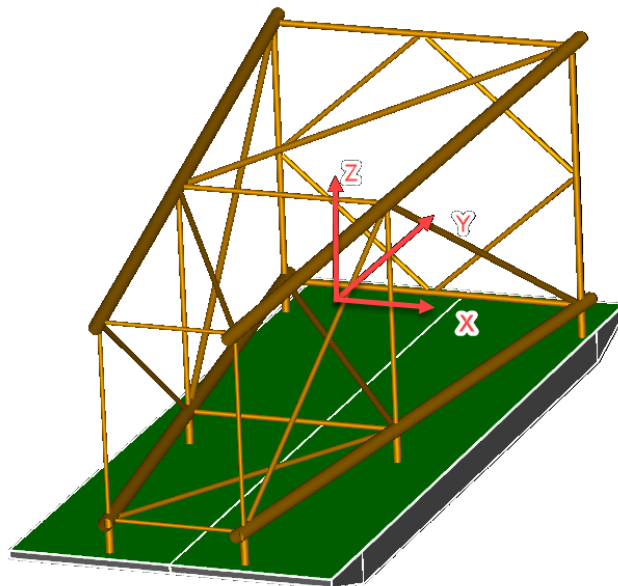
where g is the acceleration of gravity.

4 SAMPLE PROBLEMS

The structure shown in the following Figures was used in Sample Problems 1 and 2 to illustrate various capabilities of the TOW program. Sample Problem 3 is a detailed transportation analysis with all transportation bracing modeled.

1. In Sample Problem 1, six load cases were generated, each containing inertia load due to a single unit acceleration corresponding to either sway, surge, heave, pitch, roll or yaw motion. The transportation cases were accounted for by creating load combinations containing the appropriate unit load cases and load factors. Loading due to unmodeled steel is specified in load case 'MOD1' of the model file. Only tie down cans were modeled and the tie down braces were represented by a lateral support.
2. Sample Problem 2 is the same as Sample Problem 1 except that an inertia load case was generated using motion data for each of the transportation cases instead of defining combinations of unit load cases.
3. Sample Problem 3 is a detailed tow analysis with all transportation bracing modeled. In this problem, dead load or self weight is supported only by the tie down cans. The tie down diagonal bracing is effective only for inertia loading due to barge/structure motion.

The model in the examples was rotated so the global X, Y and Z axes corresponded to the barge pitch, roll and yaw axes, respectively, with the barge deck located in the global XY plane at Z elevation 0.0. See the following figure.



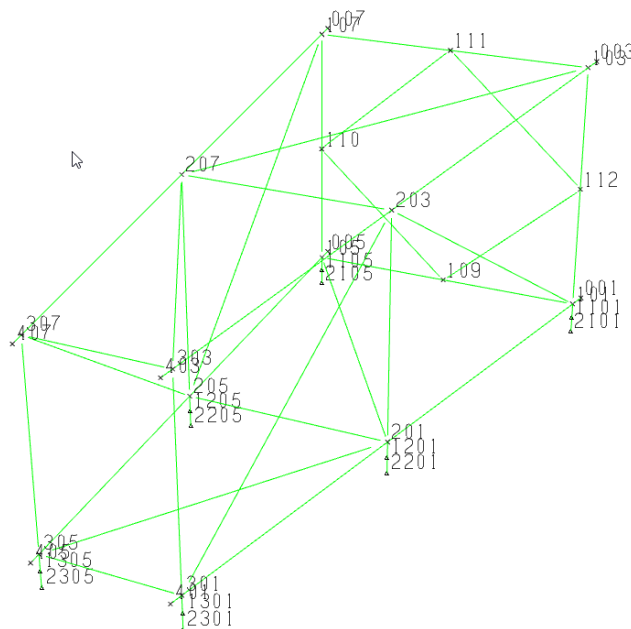
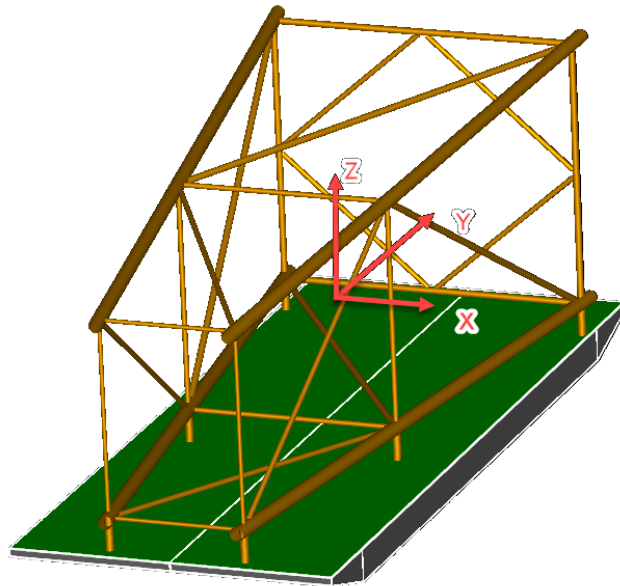
For each sample problem, the transportation environment requires that the structure be checked for motion forces due to a 12.5 degree pitch angle and a 20.0 degree roll angle, each occurring

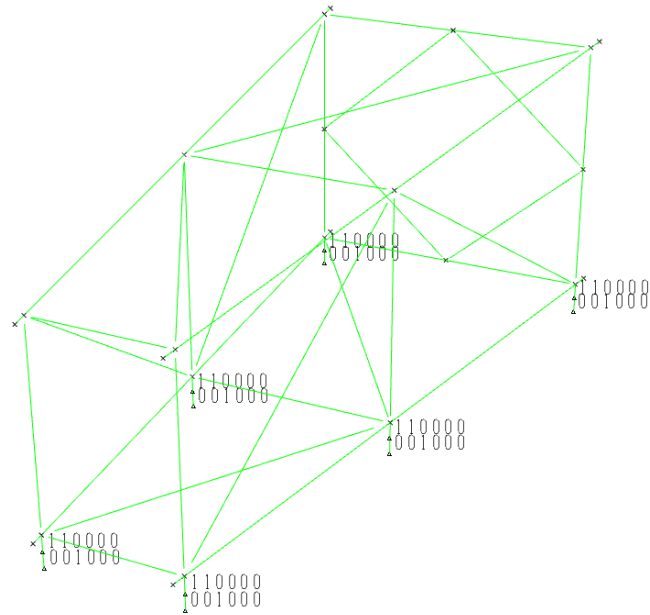
simultaneously with 0.2G heave. Pitch and roll periods were assumed to be 10 seconds. The transportation cases that were considered are as follows:

Transportation Motion	Linear (G's)			Rotation (d/s ²)		
	X	Y	Z	X	Y	Z
CW Pitch Up Hv		0.216	1.176	4.93		
CW Pitch Dwn Hv		0.216	0.776	4.93		
CCW Pitch Up Hv		-0.216	1.176	-4.93		
CW Roll Up Hv	0.34		1.14		7.89	
CW Roll Dwn Hv	0.34		0.74		7.89	
CCW Roll Up Hv	-0.34		1.14		-7.89	
CCW Roll Dwn Hv	-0.34		0.74		-7.89	

4.1 SAMPLE PROBLEM 1

In Sample Problem 1, the tie down were modeled as vertical members such that the bottom tie down joint is a vertical support (fixity=001) and the middle tie down joint is a lateral support (fixity=110) used to account for diagonal transportation bracing not included in the model.





Six unit inertia load cases were generated. Load cases 1,2 and 3 corresponded to the inertia load created by a 1G acceleration in the -X, -Y and +Z directions, respectively. Load cases 4, 5 and 6 were created for a 1 deg/sec² angular acceleration, respectively.

The weight of unmodeled items such as walkways, lifting eyes and mudmats were accounted for in load case 'MOD1' of the SACS model file.

The following is the portion of the SACS model file containing the members and joints representing the tie down and the loading corresponding to unmodeled steel:

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	OPTIONS	EN	SDUC	2 1	PTPT	PT		
3	GRUP							
4	*****Other Groups are omitted*****							
5	GRUP TIE	24.000	1.000	29.0011.2036.00	1	1.001.00	0.500	490.00
6	MEMBER							
7	*****Other Members are omitted*****							
8	MEMBER 21011101	TIE						
9	MEMBER 11011101	TIE						
10	MEMBER 21051105	TIE						
11	MEMBER 11051105	TIE						
12	MEMBER 22011201	TIE						
13	MEMBER 1201201	TIE						
14	MEMBER 22051205	TIE						
15	MEMBER 1205205	TIE						
16	MEMBER 23011301	TIE						
17	MEMBER 1301301	TIE						
18	MEMBER 23051305	TIE						
19	MEMBER 1305305	TIE						
20	JOINT							
21	*****Other Joints are omitted*****							
22	JOINT 1101	38.	164.	-4.	1.500	110000		
23	JOINT 1105	-38.	164.	-4.	-1.500	110000		
24	JOINT 1201	26.	69.	-4.	3.000	110000		
25	JOINT 1205	-26.	69.	-4.	-3.000	110000		

```

25 JOINT 1301 16. -6. -4. 9.756 -6.000 110000
26 JOINT 1305 -16. -6. -4. -9.756 -6.000 110000
27 JOINT 2101 38. 164. -8. 1.500 001000
28 JOINT 2105 -38. 164. -8. -1.500 001000
29 JOINT 2201 26. 69. -8. 3.000 001000
30 JOINT 2205 -26. 69. -8. -3.000 001000
31 JOINT 2301 16. -6. -8. 9.756 -6.000 001000
32 JOINT 2305 -16. -6. -8. -9.756 -6.000 001000
33 LOAD
34 LOADCNMOD1
35 LOAD 109 -4.0000 GLOB JOIN MUDMAT
36 LOAD 110 -4.0000 GLOB JOIN MUDMAT
37 LOAD 111 -4.0000 GLOB JOIN MUDMAT
38 LOAD 112 -4.0000 GLOB JOIN MUDMAT
39 LOAD 301 -2.0000 GLOB JOIN LIFTEYE
40 LOAD 305 -2.0000 GLOB JOIN LIFTEYE
41 LOAD 303 -2.0000 GLOB JOIN LIFTEYE
42 LOAD 307 -2.0000 GLOB JOIN LIFTEYE
43 LOAD Z 305 307 -0.1000 -0.1000 GLOB UNIF WALKWAY
44 LOAD Z 301 303 -0.1000 -0.1000 GLOB UNIF WALKWAY
45 LOAD Z 301 305 -0.1000 -0.1000 GLOB UNIF WALKWAY
46 END

```

Below is the Tow input file used to create the six transportation inertia unit load cases. A detailed description of each line of the input file follows:

```

      1      2      3      4      5      6      7      8
1 1234567890123456789012345678901234567890123456789012345678901234567890
2 TOWOPT EN MP WPOR 88.67 -8.0YXZ
3 JTNUM 403 407
4 LCFAC 1.0MOD1
5 WEIGHT 403 2.0
6 WEIGHT 407 2.0
7 WEIGHT 303 307 100.0 15.0 8.0
8 ACCL -1.0
9 ACCL -1.0
10 ACCL -1.0
11 ACCL -1.0
12 ACCL -1.0
13 END

```

Line 1. The global X, Y and Z coordinates of the center of motion of the barge structure system is designated on the TOWOPT input line in columns 31-51. The surge, sway and yaw axes are designated as Y, X and Z in columns 52-54..

Line 2. Position and acceleration for joints 403 and 407 are to be output as specified on the JTNUM input line.

Line 3. Load case 'MOD1' is to be converted to weight as designated on the LCFAC line.

Line 4-5. Additional joint weight of 2.0 to be accounted for in inertia load cases is specified on joints 403 and 407 on the first two WEIGHT input lines.

Line 6. Additional member weight is specified on member 303-307 on the third WEIGHT input line.

Line 7. The first ACCL input line designates that inertia loading due to 1.0G acceleration in the -X direction is to be created as load case 1. The loading is created in the +X direction, opposite to the applied acceleration.

Line 8. Inertia loading due to 1.0G acceleration in the -Y direction is to be created as load case 2 as designated on the second ACCL input line. The loading is created in the +Y direction, opposite to the applied acceleration.

Line 9. Load case 3 will be consist of inertia loading in the -Z direction (the direction of gravity load) as specified by 1.0 in cols. 61-70 on the third ACCL input line.

Line 10. The next ACCL input line corresponds to load case 4 and will create inertia loading due to -1.0 deg/sec² angular acceleration about the X axis. The resulting loading will produce positive rotation about the X axis.

Line 11-12. The final two ACCL input lines will create load cases 5 and 6 containing inertia loading due to -1.0 deg/sec² angular acceleration about the Y and Z axes, respectively, resulting in positive rotation about the Y and Z axes, respectively.

The following is a portion of the output listing file created by the Tow program module:

***** MEMBER GROUP REPORT *****

GRUP NO	LABEL	SEGM	SECTION ID	OUTSIDE DIAMETER IN	WALL THICK IN	WEIGHT/ LENGTH K/FT	TOTAL LENGTH FT	TOTAL WEIGHT KIPS	NUMBER MEMBERS
1	LG2	1		42.00	1.375	0.597	24.6	14.69	4
2		2		41.25	1.000	0.430	339.7	146.16	
3		3		42.00	1.375	0.597	19.6	11.70	
4	LG3	1		42.00	1.375	0.597	27.0	16.12	4
5		2		41.25	1.000	0.430	260.7	112.17	
6		3		42.00	1.375	0.597	17.4	10.39	
7	LG4	1		42.00	1.375	0.597	14.1	8.45	4
8	T01	1		16.00	0.625	0.103	725.7	74.55	12
9	T02	1		20.00	0.750	0.154	700.7	108.14	12
10	T03	1		12.75	0.500	0.065	150.5	9.85	4
11	LG1	1		42.00	1.375	0.597	20.0	11.94	4
12	TIE	1		24.00	1.000	0.246	48.0	11.80	12

***** ADDED WEIGHT DATA *****

WEIGHT NO.	WEIGHT TYPE	CONC. WEIGHT KIPS	START DISTANCE FT	***** START WT K/FT	DISTRIBUTED END WT K/FT	***** LENGTH FT	***** 1ST	***** 2ND	***** 3ND	***** 4TH	***** 5TH	***** 6TH	***** 7TH	***** 8TH
1	JOINT	4.000	0.00	0.000	0.000	0.00	109							
2	JOINT	4.000	0.00	0.000	0.000	0.00	110							
3	JOINT	4.000	0.00	0.000	0.000	0.00	111							
4	JOINT	4.000	0.00	0.000	0.000	0.00	112							
5	JOINT	2.000	0.00	0.000	0.000	0.00	301							
6	JOINT	2.000	0.00	0.000	0.000	0.00	305							
7	JOINT	2.000	0.00	0.000	0.000	0.00	303							
8	JOINT	2.000	0.00	0.000	0.000	0.00	307							
9	MEMBER	0.000	0.00	0.100	0.100	45.14	305	307						
10	MEMBER	0.000	0.00	0.100	0.100	45.14	301	303						
11	MEMBER	0.000	0.00	0.100	0.100	30.10	301	305						
12	JOINT	2.000	0.00	0.000	0.000	0.00	403							
13	JOINT	2.000	0.00	0.000	0.000	0.00	407							
14	MEMBER	0.000	8.00	0.100	0.100	15.00	303	307						

SACS MODEL ADDED WEIGHT DATA

TOTAL ADDED WEIGHT	41.54 KIPS
ADDED WT CENTER OF GRAVITY X ...	-0.02 FT
Y ...	58.84 FT
Z ...	29.03 FT
TOTAL WEIGHT	577.51 KIPS
TOTAL CENTER OF GRAVITY X	-0.03 FT
Y	86.53 FT
Z	28.46 FT

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE 1 ***

JOINT      ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403      16.38     -10.00     48.30      16.38    -98.67     56.30
407     -16.38     -10.00     48.30     -16.38    -98.67     56.30

          ***** ACCELERATIONS (G'S) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403      -1.00      0.00      0.00      -1.00      0.00      0.00
407      -1.00      0.00      0.00      -1.00      0.00      0.00

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE 2 ***

JOINT      ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403      16.38     -10.00     48.30      16.38    -98.67     56.30
407     -16.38     -10.00     48.30     -16.38    -98.67     56.30

          ***** ACCELERATIONS (G'S) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403       0.00     -1.00      0.00       0.00     -1.00      0.00
407       0.00     -1.00      0.00       0.00     -1.00      0.00

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE 3 ***

JOINT      ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403      16.38     -10.00     48.30      16.38    -98.67     56.30
407     -16.38     -10.00     48.30     -16.38    -98.67     56.30

          ***** ACCELERATIONS (G'S) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403       0.00      0.00      1.00       0.00      0.00      1.00
407       0.00      0.00      1.00       0.00      0.00      1.00

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE 4 ***

JOINT      ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403      16.38     -10.00     48.30      16.38    -98.67     56.30
407     -16.38     -10.00     48.30     -16.38    -98.67     56.30

          ***** ACCELERATIONS (G'S) ***** GLOBAL *****
          X          Y          Z          X          Y          Z
403       0.00      0.03      0.05       0.00      0.03      0.05
407       0.00      0.03      0.05       0.00      0.03      0.05

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE 5 ***															
JOINT	COORDINATES (FT)						ACCELERATIONS (G'S)								
	STRUCTURAL			GLOBAL			STRUCTURAL			GLOBAL					
	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z			
403	16.38	-10.00	48.30	16.38	-98.67	56.30	-0.03	0.00	0.01	-0.03	0.00	0.01			
407	-16.38	-10.00	48.30	-16.38	-98.67	56.30	-0.03	0.00	-0.01	-0.03	0.00	-0.01			
*** SELECTED JOINT DATA FOR INERTIA LOAD CASE 6 ***															
JOINT	COORDINATES (FT)						ACCELERATIONS (G'S)								
	STRUCTURAL			GLOBAL			STRUCTURAL			GLOBAL					
	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z			
403	16.38	-10.00	48.30	16.38	-98.67	56.30	-0.05	-0.01	0.00	-0.05	-0.01	0.00			
407	-16.38	-10.00	48.30	-16.38	-98.67	56.30	-0.05	0.01	0.00	-0.05	0.01	0.00			
* * * D Y N A M I C L O A D I N G S U M M A T I O N * * *															
(MOMENTS ABOUT SACS ORIGIN)															
(NOTE: ASTERISK BY LOAD CASE ID INDICATES STRUCTURAL WEIGHT EXCLUDED)															
LOAD CASE	* JACKET POSITION *			** ANGULAR ACCEL **			*** TRANS ACCEL ***			**** FORCE SUMMATION ****			***** MOMENT SUMMATION *****		
	PITCH	ROLL	YAW	PITCH	ROLL	YAW	SWAY	SURGE	HEAVE	SWAY	SURGE	HEAVE	PITCH	ROLL	YAW
	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z
	-----	(DEG)	-----	---	(DEG/SEC**2)	---	-----	(G)	-----	-----	(KIPS)	-----	-----	(FT-KIP)	-----
1	0.0	0.0	0.0	0.0	0.0	0.0	-1.000	0.000	0.000	577.5	0.0	0.0	0.0	16446.1	-49971.7
2	0.0	0.0	0.0	0.0	0.0	0.0	0.000	-1.000	0.000	0.0	577.5	0.0	-16446.1	0.0	-15.4
3	0.0	0.0	0.0	0.0	0.0	0.0	0.000	0.000	1.000	0.0	0.0	-577.5	-49971.7	-15.4	0.0
4	0.0	0.0	0.0	-1.0	0.0	0.0	0.000	0.000	0.000	0.0	-11.4	-0.7	1577.6	0.6	-0.5
5	0.0	0.0	0.0	0.0	-1.0	0.0	0.000	0.000	0.000	11.4	0.0	0.0	1.3	774.0	-1040.3
6	0.0	0.0	0.0	0.0	0.0	-1.0	0.000	0.000	0.000	0.7	0.0	0.0	-0.6	-32.8	1247.9

The Tow program module creates an output structural data file containing the model data and the transportation inertia loading. Six load cases pertaining to unit accelerations along the X, Y and Z axes and unit rotational accelerations about the X, Y and Z axes were created.

In this sample problem, the following transportation cases were considered:

- a. Clockwise Pitch + Gravity + Upward Heave
- b. Clockwise Pitch + Gravity + Downward Heave
- c. Counterclockwise Pitch + Gravity + Upward Heave
- d. Clockwise Roll + Gravity + Upward Heave
- e. Clockwise Roll + Gravity + Downward Heave
- f. Counterclockwise Roll + Gravity + Upward Heave
- g. Counterclockwise Roll + Gravity + Downward Heave

A load combination containing the appropriate unit load cases and load factors is required for each of the transportation cases to be considered. The load combination data may be specified directly in the Tow output structural data file or in the Post and/or Joint Can input files.

Transport Case	Load Comb	Unit Load Case Factor					
		1	2	3	4	5	6
a.	P+H1		0.216	1.176	4.93		
b.	P-H1		0.216	0.776	4.93		
c.	P+H2		-0.216	1.176	-4.93		
d.	R+H1	0.34		1.14		7.89	
e.	R-H1	0.34		0.74		7.89	
f.	R+H2	-0.34		1.14		-7.89	
g.	R-H2	-0.34		0.74		-7.89	

4.2 SAMPLE PROBLEM 2

The same transportation problem in Sample Problem 1 is addressed in the following example. However, instead of using unit inertia load cases to create the transportation cases, an individual inertia load case for each transportation case was be created by defining the motion.

Like sample problem 1, a 12.5 degree pitch angle and a 20.0 degree roll angle, each occurring simultaneously with 0.2G heave was used. Pitch and roll periods were assumed to be 10 seconds

The following transportation load cases were considered in Sample Problem 2:

Load Case	Description
P+H1	Clockwise Pitch + Gravity + Upward Heave
P-H1	Clockwise Pitch + Gravity + Downward Heave
P+H2	Pitch + Gravity + Upward Heave
R+H1	Clockwise Roll + Gravity + Upward Heave
R-H1	Clockwise Roll + Gravity + Downward Heave
R+H2	Counterclockwise Roll + Gravity + Upward Heave
R-H2	Counterclockwise Roll + Gravity + Downward Heave

Below is the Tow input file used to create the seven transportation load cases. A detailed description of the input file follows:

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	TOWOPT	EN	MP	WPOR	88.67	-8.0	YXZ	
3	JTNUM		403	407				
4	LCFAC		1.0	MOD1				
5	WEIGHT		403		2.0			
6	WEIGHT		407		2.0			
7	WEIGHT	303	307		100.0	15.0	8.0	
8	MOTIONP+H1			12.5	10.0	0.2		G
9	MOTIONP-H1			12.5	10.0	-0.2		G
10	MOTIONP+H2			-12.5	10.0	0.2		G
11	MOTIONR+H1	20.0	10.0			0.2		G
12	MOTIONR-H1	20.0	10.0			-0.2		G
13	MOTIONR+H2	20.0	10.0			0.2		G
14	MOTIONR-H2	20.0	10.0			-0.2		G
15	END							

Line 1. The TOWOPT line designates that the center of motion is 0.0, 88.67 and -8.0. The surge, sway and heave axes correspond to the global Y, X and Z axes, respectively.

Line 2. Position and acceleration for joints 403 and 407 are to be output as specified on the JTNUM input line.

Line 3. Load case MOD1 is to be converted to weight as designated on the LCFAC line.

Line 4-5. Additional joint weight of 2.0 to be accounted for in inertia load cases is specified on joints 403 and 407 on the first two WEIGHT input lines.

Line 6. Additional member weight is specified on member 303-307 on the third WEIGHT input line.

Line 7. The first MOTION line defines the inertia load case 'P+H1' as follows:

1. The pitch angle is 12.5 degrees as designated in columns 21-25.
2. The pitch period is 10.0 seconds as specified in columns 26-30.
3. The heave acceleration is designated in columns 51-55 as 0.20G.
4. The effects of the self weight of the structure are to be included in this load case as stipulated by 'G' in column 77.

Line 8. Inertia loading corresponding to transportation case B will be created as load case 'P-H1' as designated on the second MOTION line.

The following is a portion of the output listing file created by the Tow program module:

***** MEMBER GROUP REPORT *****

GRUP NO	LABEL	SEGM	SECTION ID	OUTSIDE DIAMETER IN	WALL THICK IN	WEIGHT/ LENGTH K/FT	TOTAL LENGTH FT	TOTAL WEIGHT KIPS	NUMBER MEMBERS
1	LG2	1		42.00	1.375	0.597	24.6	14.69	4
2		2		41.25	1.000	0.430	339.7	146.16	
3		3		42.00	1.375	0.597	19.6	11.70	
4	LG3	1		42.00	1.375	0.597	27.0	16.12	4
5		2		41.25	1.000	0.430	260.7	112.17	
6		3		42.00	1.375	0.597	17.4	10.39	
7	LG4	1		42.00	1.375	0.597	14.1	8.45	4
8	T01	1		16.00	0.625	0.103	725.7	74.55	12
9	T02	1		20.00	0.750	0.154	700.7	108.14	12
10	T03	1		12.75	0.500	0.065	150.5	9.85	4
11	LG1	1		42.00	1.375	0.597	20.0	11.94	4
12	TIE	1		24.00	1.000	0.246	48.0	11.80	12

***** ADDED WEIGHT DATA *****

WEIGHT NO.	WEIGHT TYPE	CONC. WEIGHT KIPS	START DISTANCE FT	***** START WT K/FT	DISTRIBUTED END WT K/FT	***** LENGTH FT	***** 1ST	***** 2ND	***** 3ND	***** 4TH	***** 5TH	***** 6TH	***** 7TH	***** 8TH
1	JOINT	4.000	0.00	0.000	0.000	0.00	109							
2	JOINT	4.000	0.00	0.000	0.000	0.00	110							
3	JOINT	4.000	0.00	0.000	0.000	0.00	111							
4	JOINT	4.000	0.00	0.000	0.000	0.00	112							
5	JOINT	2.000	0.00	0.000	0.000	0.00	301							
6	JOINT	2.000	0.00	0.000	0.000	0.00	305							
7	JOINT	2.000	0.00	0.000	0.000	0.00	303							
8	JOINT	2.000	0.00	0.000	0.000	0.00	307							
9	MEMBER	0.000	0.00	0.100	0.100	45.14	305	307						
10	MEMBER	0.000	0.00	0.100	0.100	45.14	301	303						
11	MEMBER	0.000	0.00	0.100	0.100	30.10	301	305						
12	JOINT	2.000	0.00	0.000	0.000	0.00	403							
13	JOINT	2.000	0.00	0.000	0.000	0.00	407							
14	MEMBER	0.000	8.00	0.100	0.100	15.00	303	307						

SACS MODEL ADDED WEIGHT DATA

TOTAL ADDED WEIGHT	41.54 KIPS
ADDED WT CENTER OF GRAVITY X ...	-0.02 FT
Y ...	58.84 FT
Z ...	29.03 FT
TOTAL WEIGHT	577.51 KIPS
TOTAL CENTER OF GRAVITY X	-0.03 FT
Y	86.53 FT
Z	28.46 FT


```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE P+H1 ***

JOINT          ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
              ***** STRUCTURAL ***** ACCELERATIONS (G'S) ***** GLOBAL *****
              X          Y          Z          X          Y          Z          X          Y          Z          X          Y          Z

403            16.38    -10.00    48.30      16.38   -98.67    56.30          0.00     0.37     1.44          0.00     0.37     1.44

407           -16.38    -10.00    48.30     -16.38   -98.67    56.30          0.00     0.37     1.44          0.00     0.37     1.44

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE P-H1 ***

JOINT          ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
              ***** STRUCTURAL ***** ACCELERATIONS (G'S) ***** GLOBAL *****
              X          Y          Z          X          Y          Z          X          Y          Z          X          Y          Z

403            16.38    -10.00    48.30      16.38   -98.67    56.30          0.00     0.37     1.04          0.00     0.37     1.04

407           -16.38    -10.00    48.30     -16.38   -98.67    56.30          0.00     0.37     1.04          0.00     0.37     1.04

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE P+H2 ***

JOINT          ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
              ***** STRUCTURAL ***** ACCELERATIONS (G'S) ***** GLOBAL *****
              X          Y          Z          X          Y          Z          X          Y          Z          X          Y          Z

403            16.38    -10.00    48.30      16.38   -98.67    56.30          0.00    -0.37     0.91          0.00    -0.37     0.91

407           -16.38    -10.00    48.30     -16.38   -98.67    56.30          0.00    -0.37     0.91          0.00    -0.37     0.91

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE R+H1 ***

JOINT          ***** STRUCTURAL ***** COORDINATES (FT) ***** GLOBAL *****
              ***** STRUCTURAL ***** ACCELERATIONS (G'S) ***** GLOBAL *****
              X          Y          Z          X          Y          Z          X          Y          Z          X          Y          Z

403            16.38    -10.00    48.30      16.38   -98.67    56.30         -0.58     0.00     1.21         -0.58     0.00     1.21

407           -16.38    -10.00    48.30     -16.38   -98.67    56.30         -0.58     0.00     1.07         -0.58     0.00     1.07

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE R-H1 ***

JOINT          ***** COORDINATES (FT) ***** ACCELERATIONS (G'S) *****
***** STRUCTURAL ***** GLOBAL ***** STRUCTURAL ***** GLOBAL *****
          X          Y          Z          X          Y          Z          X          Y          Z          X          Y          Z

403          16.38        -10.00        48.30          16.38        -98.67        56.30          -0.58          0.00          0.81          -0.58          0.00          0.81

407          -16.38        -10.00        48.30          -16.38        -98.67        56.30          -0.58          0.00          0.67          -0.58          0.00          0.67

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE R+H2 ***

JOINT          ***** COORDINATES (FT) ***** ACCELERATIONS (G'S) *****
***** STRUCTURAL ***** GLOBAL ***** STRUCTURAL ***** GLOBAL *****
          X          Y          Z          X          Y          Z          X          Y          Z          X          Y          Z

403          16.38        -10.00        48.30          16.38        -98.67        56.30           0.58          0.00          1.07           0.58          0.00          1.07

407          -16.38        -10.00        48.30          -16.38        -98.67        56.30           0.58          0.00          1.21           0.58          0.00          1.21

```

```

*** SELECTED JOINT DATA FOR INERTIA LOAD CASE R-H2 ***

JOINT          ***** COORDINATES (FT) ***** ACCELERATIONS (G'S) *****
***** STRUCTURAL ***** GLOBAL ***** STRUCTURAL ***** GLOBAL *****
          X          Y          Z          X          Y          Z          X          Y          Z          X          Y          Z

403          16.38        -10.00        48.30          16.38        -98.67        56.30           0.58          0.00          0.67           0.58          0.00          0.67

407          -16.38        -10.00        48.30          -16.38        -98.67        56.30           0.58          0.00          0.81           0.58          0.00          0.81

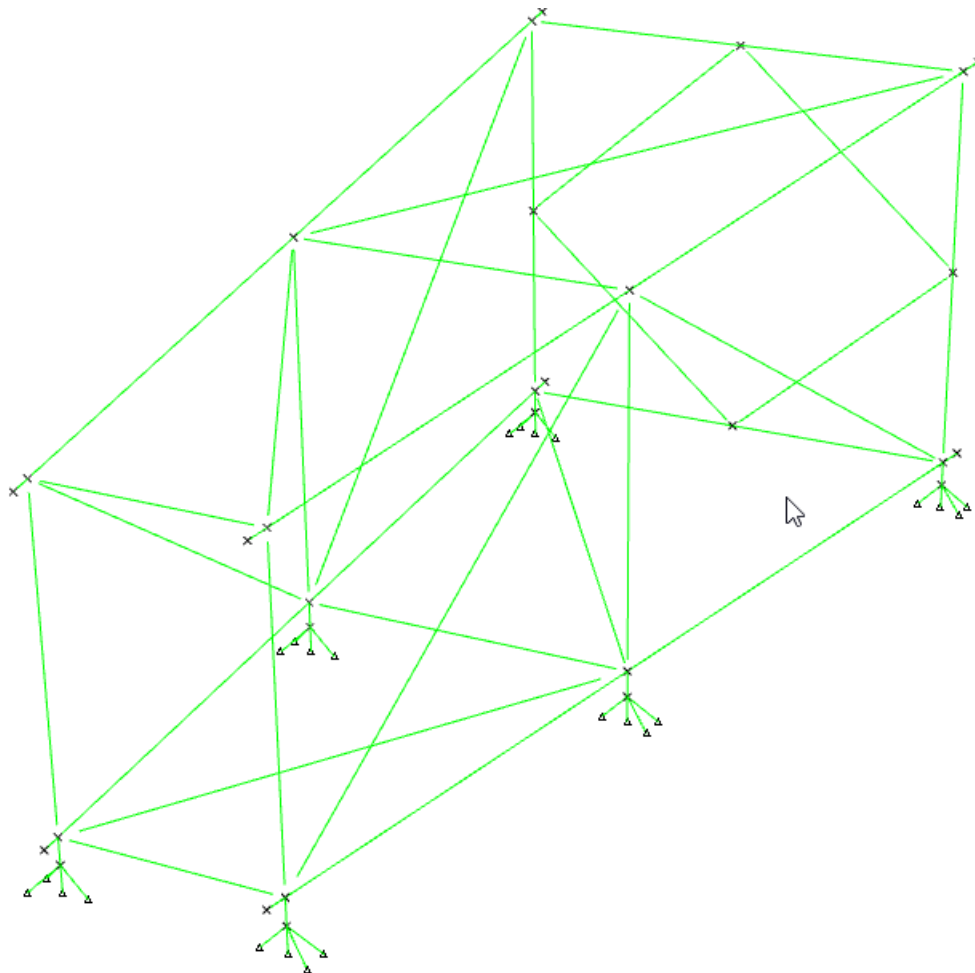
```

* * * D Y N A M I C L O A D I N G S U M M A T I O N * * *
(MOMENTS ABOUT SACS ORIGIN)
(NOTE: ASTERISK BY LOAD CASE ID INDICATES STRUCTURAL WEIGHT EXCLUDED)

LOAD CASE	* JACKET POSITION *			** ANGULAR ACCEL **			*** TRANS ACCEL ***			**** FORCE SUMMATION ****			***** MOMENT SUMMATION *****		
	PITCH	ROLL	YAW	PITCH	ROLL	YAW	SWAY	SURGE	HEAVE	SWAY	SURGE	HEAVE	PITCH	ROLL	YAW
	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z
	-----	(DEG)	-----	---	(DEG/SEC**2)	---	-----	(G)	-----	-----	(KIPS)	-----	-----	(FT-KIP)	-----
P+H1	0.0	0.0	0.0	-4.9	0.0	0.0	0.000	0.216	1.176	0.0	-181.4	-682.6	-47436.8	-15.2	0.8
P-H1	0.0	0.0	0.0	-4.9	0.0	0.0	0.000	0.216	0.776	0.0	-181.4	-451.6	-27448.0	-9.0	0.8
P+H2	0.0	0.0	0.0	4.9	0.0	0.0	0.000	-0.216	1.176	0.0	181.4	-676.0	-70126.3	-21.1	-0.8
R+H1	0.0	0.0	0.0	0.0	-7.9	0.0	-0.342	0.000	1.140	287.7	0.0	-658.1	-56941.8	11719.6	-25305.4
R-H1	0.0	0.0	0.0	0.0	-7.9	0.0	-0.342	0.000	0.740	287.7	0.0	-427.1	-36953.1	11725.8	-25305.4
R+H2	0.0	0.0	0.0	0.0	7.9	0.0	0.342	0.000	1.140	-287.7	0.0	-658.3	-56963.0	-11754.8	25305.4
R-H2	0.0	0.0	0.0	0.0	7.9	0.0	0.342	0.000	0.740	-287.7	0.0	-427.2	-36974.3	-11748.7	25305.4

4.3 SAMPLE PROBLEM 3

Sample Problem 3 is a detailed tow analysis with all transportation bracing modeled as displayed below.



Dead load or self weight is supported only by the tie down cans. The tie down diagonal bracing is effective only for inertia loading due to surge, sway, heave, pitch, roll and yaw barge/structure motion.

The transportation analysis procedure requires two separate analyses, one corresponding to the structure with all transportation bracing effective to account for motion load conditions, and the second with the tie down diagonal bracing not effective to account for the dead load condition. The two solution files will be combined to create one solution file containing both conditions. The transportation cases will be created by forming load combinations.

As in Sample Problem 1, the transportation inertia forces were created for unit inertia loads. Load cases 1,2 and 3 corresponded to the inertia load created by a 1G acceleration in the -X, -Y and +Z directions, respectively. Load cases 4, 5 and 6 were created for a 1 deg/sec² angular acceleration, respectively. The weight of unmodeled items such as walkways, lifting eyes and mudmats were accounted for in Load case 1 of the SACS model file. Below is the TOW input file, see Sample Problem 1 for a discussion of the input file and sample output listing.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	TOWOPT	EN	LD	MP	WPOR	88.67	-8.0YXZ	
3	JTNUM		403	407				
4	WEIGHT		403		2.0			
5	WEIGHT		407		2.0			
6	WEIGHT	303	307		100.0	15.0		8.0
7	ACCL					-1.0		
8	ACCL						-1.0	
9	ACCL							1.0
10	ACCL		-1.0					
11	ACCL			-1.0				
12	ACCL				-1.0			
13	END							

Note: The same procedure as Sample Problem 1 was used. In Sample Problem 1, load case 3 corresponding to inertia load in the Z direction, was used to account for the heave and the gravity component in the Z direction. In this problem, however, load case 3 is applicable to heave only because tie down diagonal braces are effective.

The OCI file created by Tow was then solved to create a common solution file containing results for the six load cases corresponding to sway, surge, heave, pitch, roll and yaw respectively.

The model file was then modified to account for the dead load condition. Below is part of the modified model file. A discussion of changes made to the original model follows:

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	OPTIONS	EN	SDUC	2	1	PTPT	PT	
3	GROUP							
4	*****other groups are omitted*****							
5	GROUP TIE	24.000	1.000	29.0011	2036.00	1	1.001.00	0.500 490.00
6	GROUP TBR	18.000	0.750	0.01	0.0136.00	1	1.001.00	0.500 1.00-2
7	MEMBER							
8	*****other members are omitted*****							
9	MEMBER	21011101	TIE					
10	MEMBER	11011101	TIE					
11	MEMBER	21051105	TIE					
12	MEMBER	11051105	TIE					
13	MEMBER	22011201	TIE					
14	MEMBER	1201201	TIE					
15	MEMBER	22051205	TIE					
16	MEMBER	1205205	TIE					
17	MEMBER	23011301	TIE					
18	MEMBER	1301301	TIE					
19	MEMBER	23051305	TIE					
20	MEMBER	1305305	TIE					
21	MEMBER	23031301	TBR					
22	MEMBER	23021301	TBR					
23	MEMBER	23041301	TBR					
24	MEMBER	22031201	TBR					
25	MEMBER	22021201	TBR					
26	MEMBER	22041201	TBR					
27	MEMBER	21031101	TBR					
28	MEMBER	21021101	TBR					
29	MEMBER	21041101	TBR					
30	MEMBER	23061305	TBR					
31	MEMBER	23071305	TBR					
32	MEMBER	23081305	TBR					
33	MEMBER	22081205	TBR					
34	MEMBER	22071205	TBR					
35	MEMBER	22061205	TBR					
36	MEMBER	21081105	TBR					
37	MEMBER	21071105	TBR					
38	MEMBER	21061105	TBR					
39	JOINT							

```

39 *****other joints are omitted*****
40 JOINT 2101 38. 164. -8. 1.500 111000
41 JOINT 2105 -38. 164. -8. -1.500 111000
42 JOINT 2201 26. 69. -8. 3.000 111000
43 JOINT 2205 -26. 69. -8. -3.000 111000
44 JOINT 2301 16. -6. -8. 9.756 -6.000 111000
45 JOINT 2305 -16. -6. -8. -9.756 -6.000 111000
46 JOINT 2302 20. -4. -8. 9.756 -6.000 111000
47 JOINT 2303 20. -8. -8. 9.756 -6.000 111000
48 JOINT 2304 12. -6. -8. 9.756 -6.000 111000
49 JOINT 2306 -12. -6. -8. -9.756 -6.000 111000
50 JOINT 2307 -20. -8. -8. -9.756 -6.000 111000
51 JOINT 2308 -20. -4. -8. -9.756 -6.000 111000
52 JOINT 2202 30. 71. -8. 3.000 111000
53 JOINT 2203 30. 67. -8. 3.000 111000
54 JOINT 2204 22. 69. -8. 3.000 111000
55 JOINT 2206 -22. 69. -8. -3.000 111000
56 JOINT 2207 -30. 67. -8. -3.000 111000
57 JOINT 2208 -30. 71. -8. -3.000 111000
58 JOINT 2102 42. 166. -8. 1.500 111000
59 JOINT 2103 42. 162. -8. 1.500 111000
60 JOINT 2104 34. 164. -8. 1.500 111000
61 JOINT 2106 -34. 164. -8. -1.500 111000
62 JOINT 2107 -42. 162. -8. -1.500 111000
63 JOINT 2108 -42. 166. -8. -1.500 111000
64 LOAD
65 LOADCN 1
66 LOAD 109 -4.0000 GLOB JOIN MUDMAT
67 LOAD 110 -4.0000 GLOB JOIN MUDMAT
68 LOAD 111 -4.0000 GLOB JOIN MUDMAT
69 LOAD 112 -4.0000 GLOB JOIN MUDMAT
70 LOAD 301 -2.0000 GLOB JOIN LIFTEYE
71 LOAD 305 -2.0000 GLOB JOIN LIFTEYE
72 LOAD 303 -2.0000 GLOB JOIN LIFTEYE
73 LOAD 307 -2.0000 GLOB JOIN LIFTEYE
74 LOAD Z 305 307 -0.1000 -0.1000 GLOB UNIF WALKWAY
75 LOAD Z 301 303 -0.1000 -0.1000 GLOB UNIF WALKWAY
76 LOAD Z 301 305 -0.1000 -0.1000 GLOB UNIF WALKWAY
77 LOAD Z 101 201 -0.59716.15000-0.5971 GLOB UNIF DEAD
78 LOAD Z 101 201 6.15000-0.430384.6893-0.4303 GLOB UNIF DEAD
79 LOAD Z 101 201 90.8393-0.59714.90000-0.5971 GLOB UNIF DEAD
80 LOAD Z 103 203 -0.59716.15000-0.5971 GLOB UNIF DEAD
81 LOAD Z 103 203 6.15000-0.430385.1595-0.4303 GLOB UNIF DEAD
82 LOAD Z 103 203 91.3095-0.59714.90000-0.5971 GLOB UNIF DEAD
83 LOAD Z 105 205 -0.59716.15000-0.5971 GLOB UNIF DEAD
84 LOAD Z 105 205 6.15000-0.430384.6893-0.4303 GLOB UNIF DEAD
85 LOAD Z 105 205 90.8393-0.59714.90000-0.5971 GLOB UNIF DEAD
86 LOAD Z 107 207 -0.59716.15000-0.5971 GLOB UNIF DEAD
87 LOAD Z 107 207 6.15000-0.430385.1595-0.4303 GLOB UNIF DEAD
88 LOAD Z 107 207 91.3095-0.59714.90000-0.5971 GLOB UNIF DEAD
89 LOAD Z 201 301 -0.59716.75000-0.5971 GLOB UNIF DEAD
90 LOAD Z 201 301 6.75000-0.430364.9875-0.4303 GLOB UNIF DEAD
91 LOAD Z 201 301 71.7375-0.59714.35000-0.5971 GLOB UNIF DEAD
92 LOAD Z 203 303 -0.59716.75000-0.5971 GLOB UNIF DEAD
93 LOAD Z 203 303 6.75000-0.430365.3612-0.4303 GLOB UNIF DEAD
94 LOAD Z 203 303 72.1112-0.59714.35000-0.5971 GLOB UNIF DEAD
95 LOAD Z 205 305 -0.59716.75000-0.5971 GLOB UNIF DEAD
96 LOAD Z 205 305 6.75000-0.430364.9875-0.4303 GLOB UNIF DEAD
97 LOAD Z 205 305 71.7375-0.59714.35000-0.5971 GLOB UNIF DEAD
98 LOAD Z 207 307 -0.59716.75000-0.5971 GLOB UNIF DEAD
99 LOAD Z 207 307 6.75000-0.430365.3612-0.4303 GLOB UNIF DEAD
100 LOAD Z 207 307 72.1112-0.59714.35000-0.5971 GLOB UNIF DEAD
101 LOAD Z 301 401 -0.5971 -0.5971 GLOB UNIF DEAD
102 LOAD Z 303 403 -0.5971 -0.5971 GLOB UNIF DEAD
103 LOAD Z 305 405 -0.5971 -0.5971 GLOB UNIF DEAD
104 LOAD Z 307 407 -0.5971 -0.5971 GLOB UNIF DEAD
105 LOAD Z 109 112 -0.1027 -0.1027 GLOB UNIF DEAD
106 LOAD Z 110 109 -0.1027 -0.1027 GLOB UNIF DEAD
107 LOAD Z 111 110 -0.1027 -0.1027 GLOB UNIF DEAD
108 LOAD Z 112 111 -0.1027 -0.1027 GLOB UNIF DEAD

```

109	LOAD Z 201 203	-0.1027	-0.1027	GLOB UNIF	DEAD
110	LOAD Z 201 205	-0.1027	-0.1027	GLOB UNIF	DEAD
111	LOAD Z 203 207	-0.1027	-0.1027	GLOB UNIF	DEAD
112	LOAD Z 203 301	-0.1027	-0.1027	GLOB UNIF	DEAD
113	LOAD Z 205 207	-0.1027	-0.1027	GLOB UNIF	DEAD
114	LOAD Z 207 303	-0.1027	-0.1027	GLOB UNIF	DEAD
115	LOAD Z 305 201	-0.1027	-0.1027	GLOB UNIF	DEAD
116	LOAD Z 307 205	-0.1027	-0.1027	GLOB UNIF	DEAD
117	LOAD Z 101 109	-0.1543	-0.1543	GLOB UNIF	DEAD
118	LOAD Z 101 112	-0.1543	-0.1543	GLOB UNIF	DEAD
119	LOAD Z 101 203	-0.1543	-0.1543	GLOB UNIF	DEAD
120	LOAD Z 103 111	-0.1543	-0.1543	GLOB UNIF	DEAD
121	LOAD Z 103 207	-0.1543	-0.1543	GLOB UNIF	DEAD
122	LOAD Z 105 110	-0.1543	-0.1543	GLOB UNIF	DEAD
123	LOAD Z 109 105	-0.1543	-0.1543	GLOB UNIF	DEAD
124	LOAD Z 110 107	-0.1543	-0.1543	GLOB UNIF	DEAD
125	LOAD Z 111 107	-0.1543	-0.1543	GLOB UNIF	DEAD
126	LOAD Z 112 103	-0.1543	-0.1543	GLOB UNIF	DEAD
127	LOAD Z 201 105	-0.1543	-0.1543	GLOB UNIF	DEAD
128	LOAD Z 205 107	-0.1543	-0.1543	GLOB UNIF	DEAD
129	LOAD Z 301 303	-0.0655	-0.0655	GLOB UNIF	DEAD
130	LOAD Z 301 305	-0.0655	-0.0655	GLOB UNIF	DEAD
131	LOAD Z 303 307	-0.0655	-0.0655	GLOB UNIF	DEAD
132	LOAD Z 305 307	-0.0655	-0.0655	GLOB UNIF	DEAD
133	LOAD Z 001 101	-0.5971	-0.5971	GLOB UNIF	DEAD
134	LOAD Z 005 105	-0.5971	-0.5971	GLOB UNIF	DEAD
135	LOAD Z 003 103	-0.5971	-0.5971	GLOB UNIF	DEAD
136	LOAD Z 007 107	-0.5971	-0.5971	GLOB UNIF	DEAD
137	END				

For the dead load or self weight condition, the following changes were made to the model used for the transportation inertia load conditions:

A. The effects of the tie down diagonal braces were negated by changing the modulus of elasticity and the weight density of member group TBR to 0.01 on the appropriate GRUP input line.

B. The dead load of the modeled structure was added to load case 1 which already contained loading representing self weight of unmodeled items.

Load case 1 was then solved creating a common solution file containing results for the structures dead weight.

The Combine program module was used to consolidate the inertia load case results and the dead weight results into one solution file. The following is the Combine input file followed by a detailed description:

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	CMBOPT							
3	LCOND	LIN	1.0	LOAD CASE 1= LIN X INERTIA IN FILE 1				
4	COMP P	1	1.0					
5	LCOND	LIN	1.0	LOAD CASE 2= LIN Y INERTIA IN FILE 1				
6	COMP P	2	1.0					
7	LCOND	LIN	1.0	LOAD CASE 3= LIN Z INERTIA IN FILE 1				
8	COMP P	3	1.0					
9	LCOND	LIN	1.0	LOAD CASE 4= ROT X INERTIA IN FILE 1				
10	COMP P	4	1.0					
11	LCOND	LIN	1.0	LOAD CASE 5= ROT Y INERTIA IN FILE 1				
12	COMP P	5	1.0					
13	LCOND	LIN	1.0	LOAD CASE 6= ROT Z INERTIA IN FILE 1				
14	COMP P	6	1.0					
15	LCOND	LIN	1.0	LOAD CASE 7= DEAD W/O DIAG TIE IN 2				
16	COMP S	1	1.0					

Line 1. The first input line is the Combine options input line.

Line 2. The first LCOND input line creates load case 1 in the new solution file. By default, the type of combination will be linear (columns 7-14 blank). A comment describing the load case is in columns 25-60.

Line 3. COMP input lines define the components of the load case to be created. Load case 1 will consist of load case 1 of the primary solution file factored by 1.0 as specified by 'P' in column 6, 1 in column 10 and 1.0 in columns 11-22, respectively.

Line 4. The next set of LCOND and COMP input lines pertain to the second load case to be created. Load case 2 will consist of load case 2 from the primary solution file.

Line 14. The last load condition, load case 7, created will consist of load case 1 of the secondary solution file factored by 1.0 as specified on the COMP input line by 'S' in column 6, 1 in column 10 and 1.0 in columns 11-22, respectively.

Note: The solution file pertaining to the inertia cases should be designated as the primary solution file because it has the actual properties of the tie down diagonal braces.

The combined effects of motion forces and gravity were accounted for by creating load combinations for the following transportation cases:

- Clockwise Pitch + Gravity + Upward Heave
- Clockwise Pitch + Gravity + Downward Heave
- Counterclockwise Pitch + Gravity + Upward Heave
- Clockwise Roll + Gravity + Upward Heave
- Clockwise Roll + Gravity + Downward Heave
- Counterclockwise Roll + Gravity + Upward Heave
- Counterclockwise Roll + Gravity + Downward Heave

The following load combinations were specified in the Post and Joint Can input files:

Transport Case	Load Comb	Unit Load Case Factor						
		1	2	3	4	5	6	7
a.	P+H1		0.216	0.176	4.93			1.0
b.	P-H1		0.216	-0.224	4.93			1.0
c.	P+H2		-0.216	0.176	-4.93			1.0
d.	R+H1	0.34		0.14		7.89		1.0
e.	R-H1	0.34		-0.26		7.89		1.0
f.	R+H2	-0.34		0.14		-7.89		1.0
g.	R-H2	-0.34		-0.26		-7.89		1.0

5 INPUT LINES

ACCELERATION INPUT LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY THE ANGULAR AND TRANSLATIONAL COMPONENTS OF THE MODEL'S ACCELERATION ABOUT THE GLOBAL AXES. THE ANGULAR VELOCITIES ARE USED IN CALCULATING THE CENTRIPETAL ACCELERATIONS. SEASTATE GENERATES STRUCTURAL LOADS FROM THESE ACCELERATION COMPONENTS IN THE OPPOSITE DIRECTION TO THE IMPOSED ACCELERATION (THE SO-CALLED REVERSE EFFECTIVE FORCES OR D'ALEMBERT FORCES). THE LOADS ARE CALCULATED BASED ON THE MOST RECENTLY ENCOUNTERED CENTER LOCATION. (1- 5) ENTER 'ACCEL'. (6- 9) ENTER THE LOAD CASE NAME FOR THIS LOAD CASE. IF LEFT BLANK, PROGRAM USES LOAD CASE NAMES IN SEQUENCE OF THE LOAD CASES AS GENERATED. (10-30) ENTER THE TRANSLATION ACCELERATION COMPONENTS ABOUT THE GLOBAL X, Y AND Z AXES. (31-51) ENTER THE ROTATIONAL ACCELERATION COMPONENTS IN THE GLOBAL X, Y AND Z DIRECTIONS. POSITIVE COMPONENTS ABOUT AN AXIS ARE GIVEN BY THE RIGHT-HAND RULE. (52-72) ENTER THE ROTATIONAL VELOCITY COMPONENTS IN THE GLOBAL X, Y AND Z DIRECTIONS. POSITIVE COMPONENTS ABOUT AN AXIS ARE GIVEN BY THE RIGHT-HAND RULE. (75) ENTER 'N' IF THE STRUCTURAL WEIGHT IS NOT TO BE INCLUDED FOR THIS ACCELERATION. (77-80) ENTER THE SEASTATE 'CENTER' IDENTIFIER FOR THIS LOAD CASE.

LINE LABEL	LOAD CASE NAME	TRANSLATIONAL ACCELERATIONS GLOBAL COORDINATES			ROTATIONAL ACCELERATIONS GLOBAL COORDINATES			ROTATIONAL VELOCITIES GLOBAL COORDINATES			EXCLUDE STRUCTURAL WEIGHT OPTION	CENTER ID
		X	Y	Z	X	Y	Z	X	Y	Z		
ACCEL												
1-- 5	6--> 9	10<--16	17<--23	24<--30	31<--37	38<--44	45<--51	52<--58	59<--65	66<--72	75	77--80
DEFAULT												
ENGLISH		G'S	G'S	G'S	RAD/SEC**2	RAD/SEC**2	RAD/SEC**2	RAD/SEC	RAD/SEC	RAD/SEC		
METRIC		G'S	G'S	G'S	RAD/SEC**2	RAD/SEC**2	RAD/SEC**2	RAD/SEC	RAD/SEC	RAD/SEC		

ACCELERATION INPUT LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY THE ANGULAR AND TRANSLATIONAL COMPONENTS OF THE CARGO'S ACCELERATION ABOUT THE GLOBAL AXES. THE PROGRAM GENERATES STRUCTURAL LOADS FROM THESE ACCELERATION COMPONENTS IN THE OPPOSITE DIRECTION TO THE IMPOSED ACCELERATION (THE SO-CALLED REVERSE EFFECTIVE FORCES OR D'ALEMBERT FORCES). THE LOADS ARE CALCULATED BASED ON THE MOST RECENTLY ENCOUNTERED CENTER, POSITION AND WEIGHT LINES IN THE INPUT DECK. GRAVITY IS NOT AUTOMATICALLY INCLUDED. IF DEAD WEIGHTS ARE DESIRED THE USER MAY SPECIFY A ONE G VERTICAL ACCELERATION (GLOBAL +Z DIRECTION) IN ADDITION TO ANY OTHER SPECIFIED ACCELERATIONS.
(1- 4)	ENTER 'ACCL'.
(7-10)	ENTER THE LOAD CASE NAME FOR THIS LOAD CASE. IF LEFT BLANK, PROGRAM USES LOAD CASE NAMES IN SEQUENCE OF THE LOAD CASES AS GENERATED.
(11-40)	ENTER THE ANGULAR ACCELERATION COMPONENTS ABOUT THE GLOBAL X, Y AND Z AXES. POSITIVE COMPONENTS ABOUT AN AXIS ARE GIVEN BY THE RIGHT-HAND RULE.
(41-70)	ENTER THE TRANSLATIONAL ACCELERATION COMPONENTS IN THE GLOBAL X, Y AND Z DIRECTIONS.

LINE LABEL	LOAD CASE NAME	ANGULAR ACCELERATIONS			TRANSLATIONAL ACCELERATIONS		
		ABOUT GLOBAL X AXES	ABOUT GLOBAL Y AXES	ABOUT GLOBAL Z AXES	IN GLOBAL X DIRECT.	IN GLOBAL Y DIRECT.	IN GLOBAL Z DIRECT.
ACCL							
1-- 4	7-->10	11<--20	21<--30	31<--40	41<--50	51<--60	61<--70
DEFAULT							
ENGLISH		DEG/SEC**2	DEG/SEC**2	DEG/SEC**2	G'S	G'S	G'S
METRIC		DEG/SEC**2	DEG/SEC**2	DEG/SEC**2	G'S	G'S	G'S

ALLOWABLE STRESS MODIFIER/MATERIAL FACTOR

COLUMNS	COMMENTARY
GENERAL	FOR AISC/API WORKING STRESS CODE FORMULAS, THE 'AMOD' LINE ALLOW THE USER TO MODIFY, FOR ANY LOAD CONDITION OR LOAD COMBINATION, THE ALLOWABLE STRESSES FOR CODE CHECKING PURPOSES. THE ALLOWABLE STRESSES CALCULATED ARE FACTORED BY THE VALUE SPECIFIED. FOR NPD CODE, THIS LINE IS USED TO SPECIFY THE MATERIAL FACTOR USED FOR ALL LOAD CASES. ONLY ONE MATERIAL FACTOR MAY BE SPECIFIED AND MUST BE DESIGNATED FOR LOAD CONDITION 1. THE SPECIFIED FACTOR WILL BE USED FOR ALL SUBSEQUENT LOAD CASES.
(1- 4)	ENTER 'AMOD' ON EACH LINE OF THIS SET. THE FIRST LINE IN THIS SET SHOULD CONTAIN THE WORD 'AMOD' AS A HEADER.
(8-11)	ENTER THE LOAD CONDITION OR LOAD COMBINATION NAME IN WHICH THE ALLOWABLE STRESSES WILL BE MODIFIED. MODIFICATION OF BASIC LOAD CONDITIONS DOES NOT AFFECT ANY LOAD COMBINATION USING THOSE BASIC LOAD CONDITIONS. FOR NPD CODE, ENTER LOAD CONDITION 1.
(13-17)	ENTER THE FACTOR THE ALLOWABLE STRESS CALCULATED BY THE PROGRAM IS TO BE MULTIPLIED BY. FOR EXAMPLE A ONE-THIRD INCREASE IN ALLOWABLE STRESS IS INPUT AS 1.333. FOR NPD CODE, ENTER THE MATERIAL FACTOR TO BE USED FOR ALL LOAD CASES.
(18-77)	FOR AISC/API WORKING STRESS DESIGN, ENTER THE LOAD CASE NAMES AND THE APPROPRIATE ALLOWABLE STRESS FACTOR FOR EACH LOAD CASE DESIRED. THE INPUT DATA IN THIS LINE TERMINATES WHEN A BLANK FIELD IS READ. THESE FIELDS SHOULD BE LEFT BLANK FOR NPD CODE.

LINE LABEL	FIRST LOAD CASE		SECOND LOAD CASE		THIRD LOAD CASE		FOURTH LOAD CASE		FIFTH LOAD CASE		SIXTH LOAD CASE		SEVENTH LOAD CASE	
	LOAD CONDITION NAME	ALLOWABLE STRESS FACTOR	LOAD CONDITION NAME	ALLOWABLE STRESS FACTOR	LOAD CONDITION NAME	ALLOWABLE STRESS FACTOR	LOAD CONDITION NAME	ALLOWABLE STRESS FACTOR	LOAD CONDITION NAME	ALLOWABLE STRESS FACTOR	LOAD CONDITION NAME	ALLOWABLE STRESS FACTOR	LOAD CONDITION NAME	ALLOWABLE STRESS FACTOR
AMOD														
1-- 4	8-->11	13<--17	18-->21	23<--27	28-->31	33<--37	38-->41	43<--47	48-->51	53<--57	58-->61	63<--67	68-->71	73<--77

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS OPTIONAL AND SIGNALS THE END OF THE CALCULATIONS.

LINE LABEL	LEAVE BLANK
END	
1-- 3	4-----80

RAO INCLUDE DATA LINE

COLUMNS	COMMENTARY
GENERAL	THE RAO INCLUDE RECORD ALLOWS THE INCLUSION OF INERTIA LOADING DUE TO WAVE RESPONSES IN SPECIFIC LOAD CASES.
(1- 6)	ENTER 'INCRAO'.
(8-14)	IF THE RAO'S ARE VELOCITY DEPENDENT, ENTER THE SPECIFIC VELOCITY HERE. IF THIS VALUE DOES NOT CORRESPOND TO A GIVEN RAO VELOCITY, THEN INTERPOLATED OR EXTRAPOLATED VALUES WILL BE USED.
(15-17)	ENTER 'FPS' IF THE ENGLISH UNITS FOR VELOCITY ARE INPUT IN FEET PER SECOND. IF LEFT BLANK, THEN KNOTS WILL BE USED.
(18)	ENTER 'N' IF THE STRUCTURAL WEIGHT IS NOT TO BE INCLUDED FOR THIS ACCELERATION. THIS OPTION IS NOT USED BY THE 'TOW' PROGRAM; FOR 'TOW', THIS OPTION IS SPECIFIED ON THE 'WAVDEF' LINE.

LINE LABEL	VELOCITY	VELOCITY UNITS OPTION	EXCLUDE STRUCTURAL WEIGHT OPTION	LEAVE BLANK
INCRAO				
1-- 6	8<--14	15--17	18	19-----80
DEFAULT				
ENGLISH	KNOT OR FT/SEC			
METRIC	M/SEC			

WEIGHT SELECTION DATA LINE

COLUMNS	COMMENTARY
GENERAL	THE WEIGHT SELECTION RECORD ALLOWS THE SELECTION OF WEIGHT GROUPS TO BE INCLUDED IN SPECIFIC LOAD CASES
(1- 6)	ENTER 'INCWGT'.
(9-12)	ENTER A FOUR CHARACTER WEIGHT GROUP. THIS GROUP MUST BE THE SAME AS ENTERED ON THE WGTNS, WGTMEM, OR WGTFF LINES.
(13-80)	ENTER ADDITIONAL WEIGHT GROUPS.

LINE LABEL	INCLUDED WEIGHT GROUP SELECTIONS																	
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH	16TH	17TH	18TH
INCWGT																		
1-- 6	9<--12	13<--16	17<--20	21<--24	25<--28	29<--32	33<--36	37<--40	41<--44	45<--48	49<--52	53<--56	57<--60	61<--64	65<--68	69<--72	73<--76	77<--80

JOINT REPORT SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT UP TO SIXTEEN JOINTS FOR WHICH OUTPUT REPORTS WILL BE ISSUED. THE OUTPUT CONSISTS OF THE JOINT LOCATIONS AND ACCELERATIONS IN BOTH STRUCTURAL AND GLOBAL COORDINATES.
(1- 5)	ENTER 'JTNUM'
(11-74)	ENTER THE JOINTS FOR WHICH OUTPUT IS DESIRED.

LINE LABEL	JOINTS SELECTED FOR OUTPUT															
	JOINT 1	JOINT 2	JOINT 3	JOINT 4	JOINT 5	JOINT 6	JOINT 7	JOINT 8	JOINT 9	JOINT 10	JOINT 11	JOINT 12	JOINT 13	JOINT 14	JOINT 15	JOINT 16
JTNUM																
1-- 5	11-->14	15-->18	19-->22	23-->26	27-->30	31-->34	35-->38	39-->42	43-->46	47-->50	51-->54	55-->58	59-->62	63-->66	67-->70	71-->74

TOW LOAD CASE FACTOR/SELECTION

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE MAY BE USED TO SPECIFY THE LOAD CASES IN THE SACS IV INPUT FILE THAT ARE TO BE CONVERTED TO INERTIA LOADING. THIS LINE CAN BE REPEATED AS OFTEN AS NECESSARY TO SELECT ANY OR ALL OF THE LOAD CASES. THIS LINE SHOULD PRECEDE THE 'ACCL' OR 'MOTION' LINES THAT ARE TO INCLUDE THESE LOAD CASES. LOAD CASES SELECTED FOR PREVIOUS 'ACCL' OR 'MOTION' LINES WILL BE REPLACED BY THESE LOAD CASES.
(11-16)	ENTER THE LOAD CASE FACTOR TO BE APPLIED WHEN CONVERTING TO INERTIA LOADING.
(17-75)	ENTER THE LOAD CASE IDENTIFIERS FOR ALL LOAD CASES TO BE SELECTED. THE LOAD CASES CAN BE IN ANY ORDER.

[illegible]

RAO IDENTIFICATION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO DEFINE RAO CHARACTERISTICS TO BE USED WITH THE RESPONSE AMPLITUDE OPERATORS TO CALCULATE RESPONSES.
(7-10)	ENTER THE LOAD CASE IDENTIFIER FOR THE FIRST GENERATED LOAD CASE. THE REMAINING LOAD CASES WILL BE INCREMENTED FROM THIS CASE. IF LEFT BLANK, THE LOAD CASES WILL BE NAMED IN ORDER OF CREATION.
(11-16)	ENTER THE WAVE DIRECTION IN DEGREES. THIS QUANTITY IS USED ONLY FOR LABELING PURPOSES FOR RAO'S CONTAINED IN THE TOW INPUT FILE. FOR RAO'S CONTAINED IN THE MODEL FILE, THE WAVE DIRECTION ENTERED IS USED TO SELECT THE RAO'S WITH THE WAVE DIRECTION NEAREST THE INPUT VALUE TO CALCULATE RESPONSES. IF THE WAVE DIRECTION IS LEFT BLANK, THEN ALL RAO'S WILL BE USED.
(17-19)	ENTER THE LOAD GENERATION OPTION AS FOLLOWS: 'R+I' - GENERATES REAL AND IMAGINARY LOAD CASES 'NPT' - GENERATES A NUMBER OF LOAD CASES FOR A COMPLETE CYCLE. FOR THE 'R+I' OPTION, EACH PERIOD/FREQUENCY IN THE RAO'S WILL GENERATE TWO LOAD CASES, REAL AND IMAGINARY. THESE LOAD CASES ARE TO BE USED IN THE FATIGUE ANALYSIS WITH THE 'R+I' OPTION ON THE 'FTLOAD' DATA RECORDS. THE 'NPT' OPTION WILL GENERATE THE NUMBER OF LOAD CASES TO COMPLETE EACH CYCLE OF EACH PERIOD/FREQUENCY IN THE RAO'S.
(20-22)	ENTER THE NUMBER OF LOAD CASES FOR THE 'NPT' OPTION.

LINE LABEL	FIRST LOAD CASE ID	WAVE DIRECTION	LOAD OPTION	NUMBER LOAD POINTS	LEAVE BLANK
LCRAO					
1-- 5	7--10	11<--16	17--19	20--22	23-----80
DEFAULT			'R+I'		
ENGLISH		DEG			
METRIC		DEG			

MOTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY ACCELERATIONS USED TO CALCULATE THE INERTIA LOADING.
(1- 6)	ENTER 'MOTION'.
(7-10)	ENTER THE LOAD CASE NAME FOR THIS LOAD CASE. IF LEFT BLANK, PROGRAM USES LOAD CASE NAMES IN SEQUENCE OF THE LOAD CASES AS GENERATED.
(11-40)	ENTER THE ROLL, PITCH AND YAW ANGLES AND PERIODS.
(41-55)	ENTER THE TRANSLATIONAL ACCELERATIONS IF APPLICABLE.
(56-76)	ENTER THE CENTER OF MOTION TO BE USED IN CALCULATING THE INERTIA FORCES FOR THIS LOAD CONDITION IF DIFFERENT FROM THE DEFAULT.
(77)	TO HAVE THE PROGRAM INCLUDE THE EFFECTS OF GRAVITY IN THE TRANSLATIONAL INERTIA FORCES: IF HEAVE IS ASSUMED NORMAL TO BARGE: ENTER 'G' TO INCLUDE THE GRAVITY COMPONENT FOR SURGE, SWAY, AND HEAVE. ENTER 'L' TO INCLUDE THE GRAVITY COMPONENT FOR ONLY SURGE AND SWAY. IF HEAVE IS ASSUMED VERTICAL: ENTER 'H' TO INCLUDE THE GRAVITY COMPONENT FOR SURGE, SWAY, AND HEAVE. ENTER 'V' TO INCLUDE THE GRAVITY COMPONENT FOR ONLY SURGE AND SWAY. NOTE: LEAVE BLANK TO EXCLUDE THE EFFECTS OF GRAVITY OR IF THE TRANSLATIONAL ACCELERATION INPUT ALREADY INCLUDE THE APPROPRIATE GRAVITY COMPONENT.
(78)	ENTER 'N' IF THE STRUCTURAL WEIGHT IS TO BE IGNORED.

LINE LABEL	LOAD CASE NAME	MOTION DATA									CENTER OF MOTION			INCLUDE GRAVITY OPTION	EXCLUDE STRUCT WEIGHT OPTION
		ROLL DATA		PITCH DATA		YAW DATA		TRANSLATIONAL ACCELERATION			X	Y	Z		
		ANGLE	PERIOD	ANGLE	PERIOD	ANGLE	PERIOD	SURGE	SWAY	HEAVE					
MOTION															
1-- 6	7-->10	11<--15	16<--20	21<--25	26<--30	31<--35	36<--40	41<--45	46<--50	51<--55	56<--62	63<--69	70<--76	77	78
DEFAULT															
ENGLISH		DEG	SEC	DEG	SEC	DEG	SEC	G'S	G'S	G'S	FT	FT	FT		
METRIC		DEG	SEC	DEG	SEC	DEG	SEC	G'S	G'S	G'S	M	M	M		

MOTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY ACCELERATIONS USED TO CALCULATE THE INERTIA LOADING.
(1- 4)	ENTER 'MOTN'.
(7-10)	ENTER THE LOAD CASE NAME FOR THIS LOAD CASE. IF LEFT BLANK, PROGRAM USES LOAD CASE NAMES IN SEQUENCE OF THE LOAD CASES AS GENERATED.
(11-52)	ENTER THE ROLL, PITCH AND YAW ANGLES AND PERIODS.
(53-73)	ENTER THE TRANSLATIONAL ACCELERATIONS IF APPLICABLE.
(74)	ENTER 'G' IF THE EFFECTS OF GRAVITY (SELF WEIGHT) ARE TO BE INCLUDED. ENTER 'L' IF THE EFFECTS OF GRAVITY ARE TO BE INCLUDED IN LATERAL FORCES ONLY.
(75)	ENTER 'N' IF THE STRUCTURAL WEIGHT IS TO BE EXCLUDED FROM THESE MOTION EFFECTS.
(77-80)	ENTER THE CENTER IDENTIFIER TO SELECT THE CENTER LOCATION FOR ROTATIONAL ACCELERATIONS AND VELOCITIES. THIS IDENTIFIER IS SPECIFIED IN THE SEASTATE INPUT FILE.

LINE LABEL	LOAD CASE NAME	MOTION DATA									INCLUDE GRAVITY OPTION	EXCLUDE STRUCTURAL WEIGHT OPTION	CENTER ID
		ROLL DATA		PITCH DATA		YAW DATA		TRANSLATIONAL ACCELERATION					
		ANGLE	PERIOD	ANGLE	PERIOD	ANGLE	PERIOD	SURGE	SWAY	HEAVE			
MOTN													
1-- 4	7-->10	11<--17	18<--24	25<--31	32<--38	39<--45	46<--52	53<--59	60<--66	67<--73	74	75	77--80
DEFAULT													
ENGLISH		DEG	SEC	DEG	SEC	DEG	SEC	G'S	G'S	G'S			
METRIC		DEG	SEC	DEG	SEC	DEG	SEC	G'S	G'S	G'S			

CARGO ORIENTATION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY HOW THE CARGO MODEL IS ORIENTED IN SPACE. THE CARGO MODEL AXES MAY BE THOUGHT OF AS BEING INITIALLY COINCIDENT WITH THE GLOBAL AXES, THEN THE CARGO MODEL IS ROTATED ABOUT THE GLOBAL AXES BY SPECIFIED AMOUNTS TO BRING IT INTO THE ORIENTATION DESIRED. THE ORDER OF ROTATIONS REQUIRED MUST BE: FIRST ABOUT THE GLOBAL X AXIS, SECOND ABOUT THE GLOBAL Y AXIS, AND THIRD ABOUT THE GLOBAL Z AXIS THE ACCOMPANYING FIGURES ILLUSTRATE VARIOUS
(1- 8)	ENTER 'POSITION'.
(11-40)	ENTER THE SPECIFIED ANGLES. POSITIVE ANGLES ARE GIVEN BY THE RIGHT-HAND RULE.

LINE LABEL	ORIENTATION ANGLES			LEAVE THIS FIELD BLANK
	ANGLE ABOUT GLOBAL X	ANGLE ABOUT GLOBAL Y	ANGLE ABOUT GLOBAL Z	
POSITION				
1-- 8	11<--20	21<--30	31<--40	41-----80
DEFAULT				
ENGLISH	DEG	DEG	DEG	
METRIC	DEG	DEG	DEG	

RIGID BODY MASS MATRIX SELECTION LINE

COLUMNS	COMMENTARY
GENERAL	THIS LINE MAY BE USED TO REQUEST THAT A RIGID BODY MASS MATRIX BE CREATED. THE MASS INCLUDES MEMBERS, PLATES, AND THE ADDED WEIGHTS.
(1- 6)	ENTER 'RBMASS'.

LINE LABEL	LEAVE BLANK
RBMASS	
1-- 6	7-----80

TITLE

COLUMNS	COMMENTARY
GENERAL (2-80)	THIS LINE MUST BE THE FIRST LINE OF THE 'TOW' INPUT DECK. ENTER ANY TITLE OR OTHER DESCRIPTIVE INFORMATION.

TITLE OR OTHER DESCRIPTIVE INFORMATION
2-----80

TOW ANALYSIS OPTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA IS USED TO SPECIFY THE OVERALL 'TOW' ANALYSIS OPTIONS AND THE GLOBAL COORDINATES OF THE CENTER OF MOTION OF THE CARGO.
(1- 6)	ENTER 'TOWOPT'.
(9-10)	SELECT FROM THE FOLLOWING: 'EN' - ENGLISH UNITS. 'MN' - METRIC UNITS WITH FORCES IN KILONEWTONS. 'ME' - METRIC UNITS WITH FORCES IN KILOGRAMS.
(11-12)	ENTER 'EC' IF AN INPUT ECHO IS DESIRED.
(13-14)	ENTER 'LD' IF ONLY THE FIRST LOAD CASE IN THE MODEL FILE IS TO BE CONVERTED TO WEIGHTS.
(15-16)	ENTER 'ND' IF THE USER INPUT MEMBER LOADS, WEIGHTS AND GENERATED INERTIA LOADS ARE TO BE OUTPUT AS JOINT LOADS.
(17-18)	ENTER 'MP' IF THE MEMBER DATA REPORT IS DESIRED.
(19-20)	ENTER 'PP' IF THE PLATE DATA REPORT IS DESIRED.
(21-22)	ENTER 'WP' IF THE WEIGHT DATA REPORT IS DESIRED.
(23-24)	ENTER 'OR' FOR SUMMATION OF LOADS ABOUT MODEL ORIGIN. 'RC' FOR SUMMATION OF LOADS ABOUT CENTER OF MOTION. 'CG' FOR SUMMATION OF LOADS ABOUT CENTER OF GRAVITY.

COLUMNS	COMMENTARY
(25-26)	ENTER 'GR' TO INCLUDE GROUT WEIGHT FOR CONCENTRIC TUBULARS.
(27-28)	ENTER 'LO' TO CONSIDER ONLY LOADING (I.E., IGNORE THE SELF WEIGHT) WHEN GENERATING INERTIA LOADING.
(29-30)	BY DEFAULT LOADING DEFINED IN THE MODEL IS NOT INCLUDED IN THE STRUCTURAL DATA FILE CREATED. ENTER 'AL' IF LOADING DEFINED IN THE MODEL IS TO BE PASSED TO THE STRUCTURAL DATA FILE CREATED.
(31-51)	ENTER THE STRUCTURAL GLOBAL COORDINATES OF THE CENTER OF MOTION. THE CARGOS ANGULAR VELOCITIES AND ACCELERATIONS ARE MEASURED ABOUT AXES THROUGH THIS POINT.
(52)	ENTER STRUCTURAL GLOBAL AXIS CORRESPONDING TO THE SURGE AXIS. THIS IS THE AXIS ABOUT WHICH ROLL OCCURS.
(53)	ENTER STRUCTURAL GLOBAL AXIS CORRESPONDING TO THE SWAY AXIS. THIS IS THE AXIS ABOUT WHICH PITCH OCCURS.
(54)	ENTER STRUCTURAL GLOBAL AXIS CORRESPONDING TO THE HEAVE AXIS. THIS IS THE AXIS ABOUT WHICH YAW OCCURS.
(56-61)	ENTER THE WEIGHT CONTINGENCY FACTOR TO BE APPLIED TO THE CALCULATED WEIGHT OF MODELED ELEMENTS.
(63-64)	ENTER 'DS' TO INCLUDE THE INERTIA LOADING FROM DUMMY

LINE LABEL	UNITS	INPUT ECHO OPTION	LOAD OPTIONS		REPORT OPTIONS				PILE GROUT OPTION	CONSIDER LOADING ONLY	PASS LOADING	CENTER OF MOTION			'TOW' COORDINATE SYSTEM			WEIGHT CONTINGENCY FACTOR	DUMMY STRUCTURE OPTION	LEAVE BLANK
			LOAD DATA OPTION	JOINT LOADS ONLY OPTION	MEMBER REPORT	PLATE REPORT	WEIGHT REPORT	LOAD SUM OPTION				X	Y	Z	SURGE & ROLL AXIS	SWAY & PITCH AXIS	HEAVE & YAW AXIS			
TOWOPT																				
1-- 6	9--10	11--12	13--14	15--16	17--18	19--20	21--22	23--24	25--26	27--28	29--30	31<--37	38<--44	45<--51	52	53	54	56--61	63--64	65--80
DEFAULT	'EN'							'OR'										1		
ENGLISH												FT	FT	FT						
METRIC												M	M	M						

WAVE DEFINITION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO DEFINE WAVE CHARACTERISTICS TO BE USED WITH THE RESPONSE AMPLITUDE OPERATORS (RAO'S) TO CALCULATE SINGLE AMPLITUDE RESPONSES.
(7-10)	ENTER THE LOAD CASE IDENTIFIER FOR THE FIRST GENERATED LOAD CASE. THE REMAINING LOAD CASES WILL BE INCREMENTED FROM THIS CASE. IF LEFT BLANK, THE LOAD CASES WILL BE NAMED IN ORDER OF CREATION.
(12-14)	ENTER THE NUMBER OF POINTS DURING THE WAVE CYCLE THAT ARE TO BE USED TO CREATE LOADS.
(15-20)	ENTER THE WAVE HEIGHT.
(21-26)	ENTER THE WAVE PERIOD. IF THIS PERIOD DOES NOT CORRESPOND TO AN RAO TABLE ENTRY, INTERPOLATION OR EXTRAPOLATION WILL BE USED.
(27-32)	ENTER THE WAVE DIRECTION IN DEGREES. THIS QUANTITY IS USED ONLY FOR LABELING PURPOSES.
(75)	ENTER 'N' IF THE STRUCTURAL WEIGHT IS NOT TO BE INCLUDED FOR THIS LOAD CASE.

LINE LABEL	FIRST LOAD CASE ID	NUMBER OF POINTS	WAVE HEIGHT	WAVE PERIOD	WAVE DIRECTION	LEAVE BLANK	EXCLUDE STRUCTURAL WEIGHT OPTION
WAVDEF							
1-- 6	7--10	12-->14	15<--20	21<--26	27<--32	33-----74	75
DEFAULT		20					
ENGLISH			FT	SEC	DEG		
METRIC			M	SEC	DEG		

ADDITIONAL JOINT WEIGHT LINES

COLUMNS	COMMENTARY
GENERAL	THIS LINE SET IS USED TO ADD CONCENTRATED WEIGHTS AT THE JOINTS OF THE STRUCTURE IN ADDITION TO ANY THAT MIGHT APPEAR IN THE SACS IV INPUT DECK. IT FOLLOWS THE POSITION LINE IF THERE IS ONE. IF THIS LINE IS USED IN ONE LOAD CASE THEN THE WEIGHT ADDED WILL REMAIN IN EFFECT IN SUBSEQUENT LOAD CASES UNLESS IT IS OVERRIDDEN BY A NEW WEIGHT LINE FOR THE SAME JOINT IN THE LATER LOAD CASE. TO REMOVE THE ADDED WEIGHT FOR A LATER LOAD CASE THIS LINE WOULD BE REPEATED IN THE LATER LOAD CASE WITH A ZERO OR A BLANK IN COLUMNS 21-30; FOR SUBSEQUENT LOAD CASES THE ADDED WEIGHT WILL NOT APPEAR UNLESS OVERRIDDEN AGAIN. THIS LINE MAY BE REPEATED AS NECESSARY TO INCLUDE WEIGHTS AT AS MANY JOINTS AS DESIRED. (1- 6) ENTER 'WEIGHT'. THE FIRST LINE OF THIS SET IS HEADER LINE HAVING ONLY THIS ENTRY. THE HEADER LINE IS REQUIRED FOR EACH LOAD CASE. ONE HEADER LINE PER LOAD CASE WILL SUFFICE FOR BOTH ADDED JOINT AND MEMBER WEIGHTS. (12-15) ENTER THE JOINT NAME WHERE THE ADDED WEIGHT IS APPLIED. (21-30) ENTER THE AMOUNT OF WEIGHT TO BE ADDED TO THIS JOINT.

LINE LABEL	JOINT NAME	WEIGHT	LEAVE THIS FIELD BLANK
WEIGHT			
1-- 6	12-->15	21<--30	31-----80
DEFAULT			
ENGLISH		KIP	
METRIC		TONNE	

ADDITIONAL JOINT WEIGHT LINES

COLUMNS

COMMENTARY

(1- 6) ENTER 'WEIGHT'. THE FIRST LINE OF THIS SET IS HEADER LINE
 HAVING ONLY THIS ENTRY. THE HEADER LINE IS REQUIRED FOR EACH
 LOAD CASE. ONE HEADER LINE PER LOAD CASE WILL SUFFICE FOR
 BOTH ADDED JOINT AND MEMBER WEIGHTS.

(12-15) ENTER FIRST JOINT NAME FOR MEMBER

(17-20) ENTER SECOND JOINT NAME FOR MEMBER

(21-30) ENTER THE AMOUNT OF CONCENTERATED WEIGHT TO BE ADDED TO THIS MEMBER.

(31-40) ENTER THE DISTRIBUTED WEIGHT FOR THIS MEMBER.

(41-50) ENTER THE LENGTH OF DISTRIBUTED WEIGHT. THE DEFAULT IS THE MEMBER LENGTH.

(61-70) ENTER THE DISTANCE FROM FIRST JOINT TO CONCENTRATED WEIGHT OR BEGINNING OF DISTRIBUTED WEIGHT.

LINE LABEL	1ST JOINT NAME	2ND JOINT NAME	CON. WEIGHT	DIST. WEIGHT	LENGTH OF DIST. WEIGHT	DISTANCE FROM FIRST JOINT TO CON. WEIGHT	LEAVE THIS FIELD BLANK
WEIGHT							
1-- 6	12-->15	17-->20	21-->30	31-->40	41-->50	61-->70	71-----80
DEFAULT					MEMBER LENGTH		
ENGLISH			KIP	LB / FT	FT	FT	
METRIC			TONNE	KG / M	M	M	